

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A168NRS20HN001\\AMMONIA STORAGE DRAIN VESSEL-7%.pvdb

Date of Analysis : Sep 17, 2026 3:16pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2300.000 mm ID
 Vessel Design Length, Tangent to Tangent 6000.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 85 °C
 Internal Design Pressure 0.350 MPa
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 1 MPa
 External Max. Allowable Working Pressure 0 MPa
 Shop Test Pressure 0.919 MPa
 Field Test Pressure 0.919 MPa
 Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -40.0 °C
 Wind Design Code No Wind Loads
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Rings	SA-516 70	...	...	K02700	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	+N+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
EH-1	0.363	0.10	1.000	2.0000	No	No
SHELL-1	0.363	0.10	1.000	2.0000	N/A	No
SHELL-2	0.363	0.10	1.000	2.0000	N/A	No
EH-2	0.363	0.10	1.000	2.0000	No	No

Liquid Level: 2000.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.682

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
1000.00	Bar 150.0 x 14.0	ST RING -1
3049.90	Bar 150.0 x 14.0	ST RING -2
5100.00	Bar 150.0 x 14.0	ST RING -3

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Type	mm	mm	mm	mm	mm	mm		
Ellipse	50.0	50.0	10.0	8.0	5.0	...	1.00	1.00
Cylinder	3050.0	3000.0	10.0	10.0	5.0	...	1.00	1.00
Cylinder	5950.0	2900.0	10.0	10.0	5.0	...	1.00	1.00
Ellipse	6000.0	50.0	10.0	8.0	5.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	8	7.89644	0.1	0.10355
20	Ring	10	7.57379	0.1	0.24854
Ring	Ring	10	8.9507	0.1	0.14064
Ring	30	10	8.95099	0.1	0.14063
30	Ring	10	8.95099	0.1	0.14063
Ring	40	10	7.36563	0.1	0.2728
40	50	8	7.89644	0.1	0.10355

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1191.67	2844.24	No Calc	No Calc
Ring	Ring	2049.9	2844.59	1391218	8706859
Ring	30	2050.1	2844.59	1753965	8706859
30	Ring	2050.1	2844.59	No Calc	No Calc
Ring	40	1091.67	2844.08	1347114	8706859
40	50	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-2''	WRC-107/537	1.059	n/a	Failed
N2-3''	WRC-297	0.841	Shell	Passed
N3-3''	WRC-297	0.841	Shell	Passed
N5-3''	WRC-297	0.841	Shell	Passed

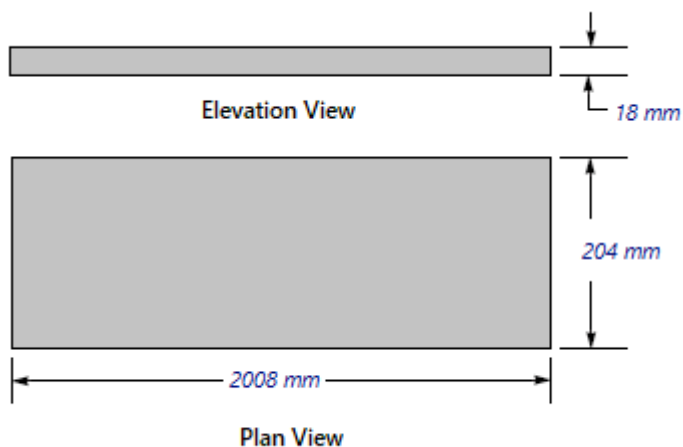
Foundation/Support Summary

Saddle Parameters:

Saddle Width	178.000 mm
Saddle Bearing Angle	130.000 deg.
Centerline Dimension	1850.000 mm
Wear Pad Width	230.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	142.000 deg.
Distance from Saddle to Tangent	750.000 mm

Baseplate Length	2008.000 mm
Baseplate Thickness	18.000 mm
Baseplate Width	204.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	13.000 mm
Height of Center Web	677.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	178465.81	N
Maximum Transverse Saddle Shear Load	12525.19	N
Maximum Longitudinal Saddle Shear Load	25050.38	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	193513.81	N
Maximum Transverse Saddle Shear Load	0.00	N
Maximum Longitudinal Saddle Shear Load	0.00	N

Weights:

Fabricated - Bare w/o removable internals	7278.1	kgm
Shop Test - Fabricated + water (full)	35379.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	7278.1	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	10118.4	kgm
Empty - Fab. + intls. + details + wghts.	10118.4	kgm
Operating - Empty + operating liquid (no ca)	27886.5	kgm
Field Test - Empty weight + test medium (full)	38220.1	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-2''	1	...	...	OK	6.80	OK	No Calc[*]
N6-2''	1	...	...	OK	6.80	OK	No Calc[*]
N2-3''	1	...	...	OK	6.80	OK	No Calc[*]
N7-2''	1	...	...	OK	5.42	OK	No Calc[*]
N4-2''	1	...	...	OK	5.42	OK	No Calc[*]
N3-3''	1	...	...	OK	6.80	OK	No Calc[*]
N5-3''	1	...	...	OK	6.80	OK	No Calc[*]
M1-24''	1	OK	...	OK	7.92	OK	Passed
N8-3''	1	...	...	OK	6.80	OK	No Calc[*]
M2-24''	1	OK	...	OK	7.94	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : M2-24''
 Minimum MAWP Shells/Flanges : 1 Element : EH-2
 Minimum MAPnc Shells/Flanges : 1 Element : EH-2

Computed Vessel M.A.W.P. : 1 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N1-2''	0.000	90.000	109.600
10	N6-2''	0.000	270.000	109.600
20	N2-3''	750.000	90.000	146.862
20	N7-2''	1250.000	90.000	69.600
20	N4-2''	2850.000	90.000	69.600
20	N3-3''	1850.000	90.000	146.862
30	N5-3''	4250.000	90.000	146.862
30	M1-24''	5450.000	90.000	1179.200
30	N8-3''	4550.000	90.000	146.862
40	M2-24''	0.000	0.000	1179.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-2''	2.000 in	Actual	LWN	2.945	12.000	...	...	220.03	600
N6-2''	2.000 in	Actual	LWN	2.945	12.000	...	...	223.48	150
N7-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	600.28	150
N4-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	600.28	150
N2-3''	3.000 in	160	WNF	3.500	11.125	150.00	10.00	600.86	300
N3-3''	3.000 in	160	WNF	3.500	11.125	150.00	10.00	600.86	300
N5-3''	3.000 in	160	WNF	3.500	11.125	150.00	10.00	600.86	300
N8-3''	3.000 in	160	WNF	3.500	11.125	150.00	10.00	600.86	150
M1-24''	24.000 in	Actual	WNF	24.000	12.000	950.00	10.00	641.13	150
M2-24''	24.000 in	Actual	WNF	24.000	12.000	950.00	8.00	330.49	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-2''	SA-333 6	8.000	10.000	...	...	...
N6-2''	SA-333 6	8.000	10.000	...	...	...
N7-2''	SA-333 6	10.000	10.000	...	...	...
N4-2''	SA-333 6	10.000	10.000	...	...	...
N2-3''	SA-333 6	10.000	10.000	8.000	10.000	...
N3-3''	SA-333 6	10.000	10.000	8.000	10.000	...
N5-3''	SA-333 6	10.000	10.000	8.000	10.000	...
N8-3''	SA-333 6	10.000	10.000	8.000	10.000	...
M1-24''	SA-516 70	10.000	10.000	8.000	10.000	...
M2-24''	SA-516 70	8.000	10.000	6.000	8.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-2''	...	90.0	200.00	0.00	EH-1
N6-2''	...	270.0	200.00	0.00	EH-1
N7-2''	1250.000	90.0	590.00	0.00	SHELL-1
N4-2''	2850.000	90.0	590.00	0.00	SHELL-1
N2-3''	750.000	90.0	590.00	0.00	SHELL-1
N3-3''	1850.000	90.0	590.00	0.00	SHELL-1
N5-3''	4250.000	90.0	590.00	0.00	SHELL-2
N8-3''	4550.000	90.0	590.00	0.00	SHELL-2
M1-24''	5450.000	90.0	590.00	0.00	SHELL-2
M2-24''	...	0.0	300.00	0.00	EH-2



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm THK X 2300.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 10.0mm THK X 2300.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 10.0mm THK X 2300.0mm ID X 2900.0mm	SA-516 70
1	PLATFORM: 1000mm X 2320mm WIDE	
1	LADDER: 2500.0mm	...
2	INSULATION: 625mm X 70mm THK	CC
2	SADDLE: 178.0mm X 130 DEG	SA-516 70
1	INSULATION: 3000mm X 70mm THK	CC
3	BAR RING STIFFENER: 2320.0mm ID X 2620.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 2900mm X 70mm THK	CC
2	CLASS 150 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
1	CLASS 150 GR 1.4, 3.0" BLIND FLANGE(S)	SA-350 LF1
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
EH-1	[10]	D	-48	-48	-29	1.000	8.000	1.00	No
EH-1	[7]	D	-48	-48	-29	0.754	10.000	1.00	No
SHELL-1	[8]	D	-48	-48	-29	0.754	10.000	1.00	No
SHELL-2	[8]	D	-48	-48	-29	0.754	10.000	1.00	No
EH-2	[10]	D	-48	-48	-29	1.000	8.000	1.00	No
EH-2	[7]	D	-48	-48	-29	0.754	10.000	1.00	No
ST RING -1	[23]	B	-29	-43	-29	0.754	10.000	1.00	No
ST RING -2	[23]	B	-29	-43	-29	0.754	10.000	1.00	No
ST RING -3	[23]	B	-29	-43	-29	0.754	10.000	1.00	No
Left Saddle	[24]	D	-48	-48	-29	0.754	10.000	1.00	No
RIGHT SADDLE	[24]	D	-48	-48	-29	0.754	10.000	1.00	No
N1-2''	[1]	D	-48	-48	-29	0.886	8.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
N6-2''	[1]	D	-48	-48	-29	0.897	8.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
N2-3''	[1]	D	-48	-48	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
N7-2''	[1]	D	-48	-48	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
N4-2''	[1]	D	-48	-48	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
N3-3''	[1]	B	-29	-43	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
N5-3''	[1]	B	-29	-43	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
M1-24''	[1]	D	-48	-48	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	D	-29	-48					
N8-3''	[1]	D	-48	-48	-29	0.740	10.000	1.00	No
Nozzle Flg	[5]	!	-29	-72					
M2-24''	[1]	D	-40	-46		0.891	8.000	1.00	No
Nozzle Flg	[5]	!	-29	-71					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -29 -40

Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -40.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code No Wind Loads

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	EH-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	2300 mm
Element Thickness	8 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	0 °C



Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	D
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Platform
Detail ID	PLATFORM
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	9120800 mm ²
Platform Weight	24444 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	2320 mm
Platform Height	1000 mm
Platform Clearance or End Offset	105 mm
Platform Force Coefficient	1.3
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	2500 mm
Unit Weight of Ladder	0.4909 N/mm
Platform Length (top head platform)	6000 mm

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2000 mm
Liquid Density	0.0006809 kg/cm ³

Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-575 mm
Height/Length of Insulation	625 mm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	N1-2''
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	86.493 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-333 6 +IT

Element From Node	10
Detail Type	Nozzle
Detail ID	N6-2''
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	2 in.



Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	68.521 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-333 6 +IT

Element From Node	10
Detail Type	Weight
Detail ID	N1-DIP PIPE
Dist. from "FROM" Node / Offset dist	-400 mm
Miscellaneous Weight	147.09 N
Offset from Element Centerline	600 mm

Element#	2/4
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	SHELL-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	2300 mm
Element Thickness	10 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	700 mm
Width of Saddle	178 mm
Height of Saddle at Bottom	1850 mm
Saddle Contact Angle	130.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	230 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	142.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2000 mm
Liquid Density	0.0006809 kg/cm³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm³

Element From Node 20
 Detail Type Ring
 Detail ID ST RING -1
 Dist. from "FROM" Node / Offset dist 950 mm
 Inside Diameter of Ring 2320 mm
 Thickness of Ring 14 mm
 Outside Diameter of Ring 2620 mm
 Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 20
 Detail Type Ring
 Detail ID ST RING -2
 Dist. from "FROM" Node / Offset dist 2999.9 mm
 Inside Diameter of Ring 2320 mm
 Thickness of Ring 14 mm
 Outside Diameter of Ring 2620 mm
 Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 20
 Detail Type Nozzle
 Detail ID N2-3''
 Dist. from "FROM" Node / Offset dist 700 mm
 Nozzle Diameter 3 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 218.16 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-333 6 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID N7-2''
 Dist. from "FROM" Node / Offset dist 1200 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 88.314 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-333 6 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID N4-2''
 Dist. from "FROM" Node / Offset dist 2800 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 88.314 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-333 6 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID N3-3''
 Dist. from "FROM" Node / Offset dist 1800 mm
 Nozzle Diameter 3 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 218.16 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-333 6 +IT

Element From Node 20
 Detail Type Weight
 Detail ID LL-1
 Dist. from "FROM" Node / Offset dist 500 mm
 Miscellaneous Weight 1000.2 N
 Offset from Element Centerline 1300 mm

Element From Node 20
 Detail Type Weight
 Detail ID DIP PIPE -2
 Dist. from "FROM" Node / Offset dist 700 mm
 Miscellaneous Weight 196.12 N
 Offset from Element Centerline 900 mm

Element# 3/4
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description SHELL-2
 Distance "FROM" to "TO" 2900 mm
 Inside Diameter 2300 mm
 Element Thickness 10 mm
 Internal Corrosion Allowance 2 mm
 Nominal Thickness 10 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.35 MPa
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0.1 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +N
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Saddle
 Detail ID RIGHT SADDLE
 Dist. from "FROM" Node / Offset dist 2200 mm
 Width of Saddle 178 mm
 Height of Saddle at Bottom 1850 mm
 Saddle Contact Angle 130.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 230 mm
 Thickness of Wear Plate 10 mm
 Contact Angle, Wear Plate (degrees) 142.0
 Friction coefficient 0.60000002
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm

Element From Node 30
 Detail Type Liquid
 Detail ID Liquid 30
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 2000 mm
 Liquid Density 0.0006809 kg/cm³

Element From Node 30
 Detail Type Insulation
 Detail ID Ins: 30



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2900	mm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³

Element From Node	30
Detail Type	Ring
Detail ID	ST RING -3
Dist. from "FROM" Node / Offset dist	2050 mm
Inside Diameter of Ring	2320 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	2620 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Nozzle
Detail ID	N5-3''
Dist. from "FROM" Node / Offset dist	1200 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	218.16 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-333 6 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	M1-24''
Dist. from "FROM" Node / Offset dist	2400 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	4379.3 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-516 70

Element From Node	30
Detail Type	Nozzle
Detail ID	N8-3''
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	251.06 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-333 6 +IT

Element From Node	30
Detail Type	Weight
Detail ID	LL-2
Dist. from "FROM" Node / Offset dist	2500 mm
Miscellaneous Weight	1000.2 N
Offset from Element Centerline	1300 mm

Element From Node	30
Detail Type	Weight
Detail ID	MH davit -1
Dist. from "FROM" Node / Offset dist	2400 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	1300 mm

Element#	4/4
Element From Node	40
Element To Node	50
Element Type	Elliptical
Description	EH-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	2300 mm
Element Thickness	8 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2000 mm
Liquid Density	0.0006809 kg/cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	625 mm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	M2-24''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	3816.1 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-516 70 +IT

Element From Node	40
Detail Type	Weight
Detail ID	MH DAVIT-2
Dist. from "FROM" Node / Offset dist	625 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	EH-1	50	...	50	...
	SHELL-1	3050	...	3000	...
	SHELL-2	5950	...	2900	...
	EH-2	6000	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	EH-1	0.3634	10	2	2300	138	1
	SHELL-1	0.3634	10	2	2300	138	2
	SHELL-2	0.3634	10	2	2300	138	3
	EH-2	0.3634	10	2	2300	138	4

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	EH-1	0.35	0.70665	0.95933	8	5.02726	1
	SHELL-1	0.35	0.94095	1.19375	10	5.03833	2
	SHELL-2	0.35	0.94097	1.19377	10	5.03827	3
	EH-2	0.35	0.70665	0.95933	8	5.02726	4

Minimum 1 1

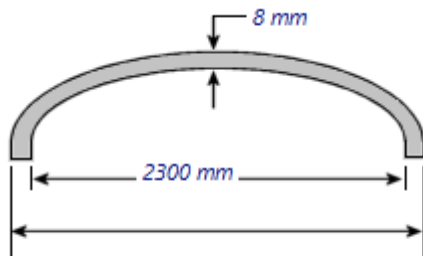
MAWP: 1 MPa, limited by: EH-2.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: EH-1

Elliptical Head Design Information:					
Design Pressure	P	0.363 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2300.000 mm	Actual Stress		69.6 MPa
Spec. Min. Thk t _e , or s _a		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		6198867.0 mm ²	SG of Contents		0.682
Element Volume		1800.7 ltr	Weight of Contents		11474.3 N
Empty Weight		5129.4 N	Operating Weight		16603.6 N
			Comp. All. Ope. Stress		77.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.363 \cdot 2304.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.363)$$

$$= 3.0273 + 2.0000 = 5.0273 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 6.0) / (0.998 \cdot 2304.0 + 0.2 \cdot 6.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.0) / (1.0 \cdot 2300.0 + 0.2 \cdot 8.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.363 \cdot (0.998 \cdot 2304.0 + 0.2 \cdot 6.0)) / (2 \cdot 1.0 \cdot 6.0)$$

= 69.645 MPa

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.363 \cdot 1152.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.363) + 2.0$$

$$= 5.038 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (1152.0 + 0.6 \cdot 8.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2304.0 / (2 \cdot 577.0))^2) / 6$$

$$= 0.997691$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.894 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.0, tr = 6.0, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 1.0$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	0.720 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-40 °C
Governing Thk.	10.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.754	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: EH-1

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	2316.00	386.00	0.0003598	35.97

$$MAEP = B / (K \cdot Do / t) = 35.9735 / (0.9 \cdot 386.0) = 0.1036 \text{ MPa}$$



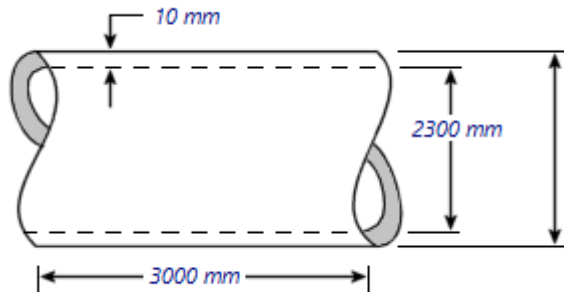
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.896	2316.00	392.78	0.0003536	35.35

$$MAEP = B/(K0*Do/t) = 35.3526/(0.9 * 392.7796) = 0.1 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.363 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2300.000 mm	Actual Stress		52.5 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		21865484.0 mm ²	SG of Contents		0.682
Element Volume		12466.5 ltr	Weight of Contents		76925.1 N
Empty Weight		16546.4 N	Operating Weight		93471.4 N
			Comp. All. Ope. Stress		96.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.363 \cdot 1152.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.363)$$

$$= 3.0383 + 2.0000 = 5.0383 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (1152.0 + 0.6 \cdot 8.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.0) / (1150.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.363 \cdot (1152.0 + 0.6 \cdot 8.0)) / (1.0 \cdot 8.0)$$

$$= 52.545 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.433 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	0.720 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-40 °C
Governing Thk.	10.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.754	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: SHELL-1

Cylindrical Shell From 20 to ST RING -1 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	1191.67	290.00	0.5136	0.0005407	54.06

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 54.0578) / (3 \cdot 290.0) = 0.2485 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.574	2320.00	1191.67	416.23	0.5136	0.0003123	31.22

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 31.2195) / (3 \cdot 416.2337) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2844.24	290.00	1.2260	0.0002177	21.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.7631) / (3 \cdot 290.0) = 0.1001 \text{ MPa}$$

Cylindrical Shell From ST RING -1 to ST RING -2 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2049.90	290.00	0.8836	0.0003060	30.59

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 30.5894) / (3 \cdot 290.0) = 0.1406 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.951	2320.00	2049.90	333.78	0.8836	0.0002504	25.04



$$MAEP = (4*B)/(3*(Do/t)) = (4*25.0353)/(3*333.7793) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2844.59	290.00	1.2261	0.0002177	21.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7604)/(3*290.0) = 0.1 \text{ MPa}$$

Cylindrical Shell From ST RING -2 to 30 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2050.10	290.00	0.8837	0.0003059	30.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.5863)/(3*290.0) = 0.1406 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.951	2320.00	2050.10	333.77	0.8837	0.0002504	25.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.0343)/(3*333.7654) = 0.1 \text{ MPa}$$

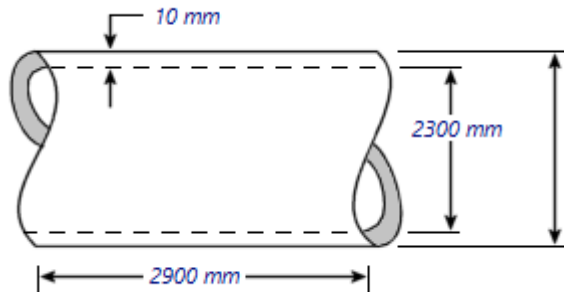
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2844.59	290.00	1.2261	0.0002177	21.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7604)/(3*290.0) = 0.1 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:			
Design Pressure P	0.363 MPa	Int. Design Temperature	85 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	2900.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	2300.000 mm	Actual Stress	52.5 MPa
Spec. Min. Thk t _e , or sa	10.0000 mm	Spec. Nominal Thk.	10.000 mm
Surface Area	21136636.0 mm ²	SG of Contents	0.682
Element Volume	12051.0 ltr	Weight of Contents	74359.9 N
Empty Weight	15994.0 N	Operating Weight	90353.8 N
		Comp. All. Ope. Stress	96.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.363 \cdot 1152.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.363)$$

$$= 3.0383 + 2.0000 = 5.0383 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (1152.0 + 0.6 \cdot 8.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.0) / (1150.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.363 \cdot (1152.0 + 0.6 \cdot 8.0)) / (1.0 \cdot 8.0)$$

$$= 52.545 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.433 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	0.720 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-40 °C
Governing Thk.	10.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.754	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cylindrical Shell From 30 to ST RING -3 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2050.10	290.00	0.8837	0.0003059	30.59

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 30.5863) / (3 \cdot 290.0) = 0.1406 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.951	2320.00	2050.10	333.77	0.8837	0.0002504	25.03

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 25.0343) / (3 \cdot 333.7654) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2844.59	290.00	1.2261	0.0002177	21.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.7604) / (3 \cdot 290.0) = 0.1 \text{ MPa}$$

Cylindrical Shell From ST RING -3 to 40 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	1091.67	290.00	0.4705	0.0005935	59.33

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 59.3344) / (3 \cdot 290.0) = 0.2728 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.366	2320.00	1091.67	432.38	0.4705	0.0003244	32.43



$$MAEP = (4*B)/(3*(Do/t)) = (4*32.4304)/(3*432.3814) = 0.1 \text{ MPa}$$

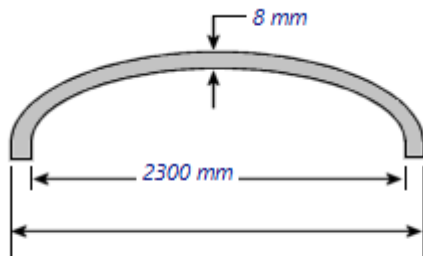
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.000	2320.00	2844.08	290.00	1.2259	0.0002177	21.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7644)/(3*290.0) = 0.1001 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: EH-2

Elliptical Head Design Information:					
Design Pressure	P	0.363 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2300.000 mm	Actual Stress	69.6 MPa	
Spec. Min. Thk t _e , or sa		8.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		6198867.0 mm ²	SG of Contents	0.682	
Element Volume		1800.7 ltr	Weight of Contents	11474.3 N	
Empty Weight		5129.4 N	Operating Weight	16603.6 N	
			Comp. All. Ope. Stress	96.7 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.363 \cdot 2304.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.363)$$

$$= 3.0273 + 2.0000 = 5.0273 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 6.0) / (0.998 \cdot 2304.0 + 0.2 \cdot 6.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.0) / (1.0 \cdot 2300.0 + 0.2 \cdot 8.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.363 \cdot (0.998 \cdot 2304.0 + 0.2 \cdot 6.0)) / (2 \cdot 1.0 \cdot 6.0)$$

= 69.645 MPa

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.363 \cdot 1152.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.363) + 2.0$$

$$= 5.038 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (1152.0 + 0.6 \cdot 8.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2304.0 / (2 \cdot 577.0))^2) / 6$$

$$= 0.997691$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 1.894 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.0, tr = 6.0, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 1.0$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	0.720 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-40 °C
Governing Thk.	10.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.754	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: EH-2

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	2316.00	386.00	0.0003598	35.97

$$\text{MAEP} = B / (K \cdot Do / t) = 35.9735 / (0.9 \cdot 386.0) = 0.1036 \text{ MPa}$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.896	2316.00	392.78	0.0003536	35.35

$$MAEP = B/(K0*Do/t) = 35.3526/(0.9 * 392.7796) = 0.1 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.919 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.225 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.777 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	0.707 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.707 • 1.0
 = 0.919 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	135.4	234.0	0.579	0.94
SHELL-1	108.8	234.0	0.465	0.94
SHELL-2	108.8	234.0	0.465	0.94
EH-2	135.4	234.0	0.579	0.94

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-2''	Nozzle	118.00	118.00	1.000
N6-2''	Nozzle	118.00	118.00	1.000
N2-3''	Nozzle	118.00	118.00	1.000
N2-3''	Pad	138.00	138.00	1.000
N7-2''	Nozzle	118.00	118.00	1.000
N4-2''	Nozzle	118.00	118.00	1.000
N3-3''	Nozzle	118.00	118.00	1.000
N3-3''	Pad	138.00	138.00	1.000
N5-3''	Nozzle	118.00	118.00	1.000
N5-3''	Pad	138.00	138.00	1.000
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
N8-3''	Nozzle	118.00	118.00	1.000
N8-3''	Pad	138.00	138.00	1.000
M2-24''	Nozzle	138.00	138.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
ST RING -1	138.00	138.00	1.000
ST RING -2	138.00	138.00	1.000
ST RING -3	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000



SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1-2''	2.56	216.00	0.012
N6-2''	2.56	216.00	0.012
N2-3''	3.92	216.00	0.018
N7-2''	2.01	216.00	0.009
N4-2''	2.01	216.00	0.009
N3-3''	3.92	216.00	0.018
N5-3''	3.92	216.00	0.018
M1-24''	23.53	234.00	0.101
N8-3''	3.92	216.00	0.018
M2-24''	23.53	234.00	0.101

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.919 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.225 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.777 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	0.707 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.707 • 1.0
 = 0.919 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	180.4	234.0	0.771	0.94
SHELL-1	136.1	234.0	0.582	0.94
SHELL-2	136.1	234.0	0.582	0.94
EH-2	180.4	234.0	0.771	0.94

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-2''	Nozzle	118.00	118.00	1.000
N6-2''	Nozzle	118.00	118.00	1.000
N2-3''	Nozzle	118.00	118.00	1.000
N2-3''	Pad	138.00	138.00	1.000
N7-2''	Nozzle	118.00	118.00	1.000
N4-2''	Nozzle	118.00	118.00	1.000
N3-3''	Nozzle	118.00	118.00	1.000
N3-3''	Pad	138.00	138.00	1.000
N5-3''	Nozzle	118.00	118.00	1.000
N5-3''	Pad	138.00	138.00	1.000
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
N8-3''	Nozzle	118.00	118.00	1.000
N8-3''	Pad	138.00	138.00	1.000
M2-24''	Nozzle	138.00	138.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
ST RING -1	138.00	138.00	1.000
ST RING -2	138.00	138.00	1.000
ST RING -3	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000



SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1-2''	2.56	216.00	0.012
N6-2''	2.56	216.00	0.012
N2-3''	3.92	216.00	0.018
N7-2''	2.01	216.00	0.009
N4-2''	2.01	216.00	0.009
N3-3''	3.92	216.00	0.018
N5-3''	3.92	216.00	0.018
M1-24''	23.53	234.00	0.101
N8-3''	3.92	216.00	0.018
M2-24''	23.53	234.00	0.101

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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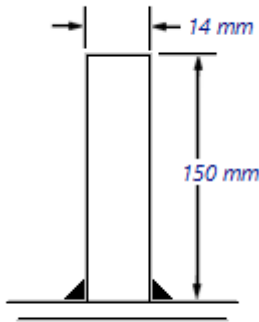
MDMT Calculations for: ST RING -1, Shell to Ring Junction, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 10.0$, $t_r = 6.03$, $c = 2.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

Stiffening Ring Calculations for: ST RING -1, SA-516 70 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 150 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1198.869	4.0000	4795.476
Ring :	2099.998	82.9999	174299.688
Total:	3298.867		179095.156

Centroid of Ring plus Shell: 54 mm

	Inertia	Distance	A*Dist ²
Shell:	6393.960	50.2899	3032028.500
Ring :	3937485.000	-28.7100	1730956.250
Total:	3943879.000		4762985.000

Available Moment of Inertia, Ring + Shell: 0.871E+07 mm⁴

Required Stress in Ring plus Shell Breq 18.72 MPa
 Required Strain in Ring plus Shell Areq 0.0001870

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (2320.0^2 \cdot 1620.7833(8.0 + 2099.998/1620.7833)0.000187) / 10.9$$

$$= 1391219 \text{ mm}^4$$

MDMT Calculations for: ST RING -2, Shell to Ring Junction, min(Curve:B, Curve:D)

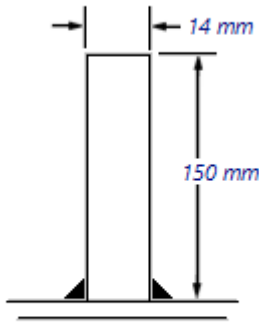
Govrn. thk, $t_g = 10.0$, $t_r = 6.03$, $c = 2.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

Stiffening Ring Calculations for: ST RING -2, SA-516 70 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 150 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1198.869	4.0000	4795.476
Ring :	2099.998	82.9999	174299.688
Total:	3298.867		179095.156

Centroid of Ring plus Shell: 54 mm

	Inertia	Distance	A*Dist ²
Shell:	6393.960	50.2899	3032028.500
Ring :	3937485.000	-28.7100	1730956.250
Total:	3943879.000		4762985.000

Available Moment of Inertia, Ring + Shell: 0.871E+07 mm⁴

Required Stress in Ring plus Shell Breq 19.28 MPa
 Required Strain in Ring plus Shell Areq 0.0001920

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (2320.0^2 \cdot 2050.0(8.0 + 2099.998/2050.0)0.000192) / 10.9$$

$$= 1753965 \text{ mm}^4$$

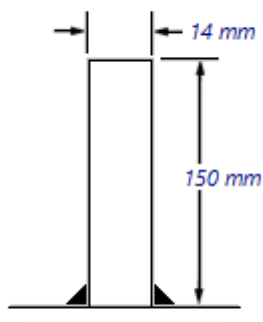
MDMT Calculations for: ST RING -3, Shell to Ring Junction, min(Curve:B, Curve:D)

Govern. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.754, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

Stiffening Ring Calculations for: ST RING -3, SA-516 70 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 150 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1198.869	4.0000	4795.476
Ring :	2099.998	82.9999	174299.688
Total:	3298.867		179095.156

Centroid of Ring plus Shell: 54 mm

	Inertia	Distance	A*Dist ²
Shell:	6393.960	50.2899	3032028.500
Ring :	3937485.000	-28.7100	1730956.250
Total:	3943879.000		4762985.000

Available Moment of Inertia, Ring + Shell: 0.871E+07 mm⁴

Required Stress in Ring plus Shell Breq 18.64 MPa
 Required Strain in Ring plus Shell Areq 0.0001860

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (2320.0^2 \cdot 1570.8834(8.0 + 2099.998/1570.8834)0.000186) / 10.9$$

$$= 1347114 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	2316	6	0.00035982	35.9735
20	Ring	1191.67	2320	8	0.0005407	54.0578
Ring	Ring	2049.9	2320	8	0.00030596	30.5894
Ring	30	2050.1	2320	8	0.00030593	30.5863
30	Ring	2050.1	2320	8	0.00030593	30.5863
Ring	40	1091.67	2320	8	0.00059348	59.3344
40	50	No Calc	2316	6	0.00035982	35.9735

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	8	7.89644	0.1	0.10355
20	Ring	10	7.57379	0.1	0.24854



Ring	Ring	10	8.9507	0.1	0.14064
Ring	30	10	8.95099	0.1	0.14063
30	Ring	10	8.95099	0.1	0.14063
Ring	40	10	7.36563	0.1	0.2728
40	50	8	7.89644	0.1	0.10355

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1191.67	2844.24	No Calc	No Calc
Ring	Ring	2049.9	2844.59	1391218	8706859
Ring	30	2050.1	2844.59	1753965	8706859
30	Ring	2050.1	2844.59	No Calc	No Calc
Ring	40	1091.67	2844.08	1347114	8706859
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	523.084	1800.72	418.467	1809.77	36.6159
20	30	1687.37	12466.5	1351.08	12509.9	118.116
30	40	1631.04	12051	1305.96	12092.9	114.173
40	50	523.084	1800.72	418.467	1809.77	36.6159
Total		4364	28118.89	3493	28222.34	305

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	2667.26	3000	1234.2	...	PLATFORM
10	Liqd	1170.13	-191.667	150	...	Liquid 10
10	Insl	24.4498	-262.5	...	...	Ins: 10
10	Nozl	9.43786	-490.535	600	...	N1-2''
10	Nozl	7.47683	-435.89	-750	...	N6-2''
10	Wght	15	-435.89	600	...	N1-DIP PIPE
20	Sadl	417.157	700	1490	...	Left Saddle
20	Liqd	7844.7	1500	150	...	Liquid 20
20	Insl	63.1445	1500	...	...	Ins: 20
20	Ring	135.139	950	...	...	ST RING -1
20	Ring	135.139	2999.9	...	...	ST RING -2
20	Nozl	23.8055	700	1194.45	...	N2-3''
20	Nozl	9.63654	1200	1175.4	...	N7-2''
20	Nozl	9.63654	2800	1175.4	...	N4-2''
20	Nozl	23.8055	1800	1194.45	...	N3-3''
20	Wght	102	500	1125.83	650	LL-1
20	Wght	20.0004	700	900	...	DIP PIPE -2
30	Sadl	417.136	2200	1490	...	RIGHT SADDLE
30	Liqd	7583.1	1450	150	...	Liquid 30
30	Insl	61.0396	1450	...	...	Ins: 30
30	Ring	135.139	2050	...	...	ST RING -3
30	Nozl	23.8055	1200	1194.45	...	N5-3''
30	Nozl	477.852	2400	1454.8	...	M1-24''
30	Nozl	27.3945	1500	1194.45	...	N8-3''
30	Wght	102	2500	1125.83	650	LL-2
30	Wght	50	2400	1300	...	MH davit -1
40	Liqd	1170.13	241.667	150	...	Liquid 40
40	Insl	24.4498	312.5	...	...	Ins: 40
40	Nozl	416.406	625	...	...	M2-24''
40	Wght	50	625	...	...	MH DAVIT-2

Total Weight of Each Detail Type:

Saddles	834.3
Platforms	2667.3
Liquid	17768.1
Insulation	173.1
Stiffeners	405.4
Nozzles	1029.3
Weights	339.0
Sum of the Detail Weights	23216.4 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	4670.1	4670.1	4670.1	4670.1	4670.1	4670.1
Saddles	834.3	834.3	834.3	834.3	834.3	834.3



Stif. Rings	405.4	405.4	405.4	405.4	405.4	405.4
Nozzles	1029.3	1029.3	1029.3	1029.3	1029.3	1029.3
Wld Weights	339.0	339.0	339.0	339.0	339.0	339.0
Platforms	...	...	...	2667.3	2667.3	2667.3
Insulation	...	...	...	173.1	173.1	173.1
Ope. Liquid	...	...	...	...	...	17768.1
Test Liquid	...	28101.7	...	...	...	...
Totals	7278.1	35379.8	7278.1	10118.4	10118.4	27886.5

Field Installation Options:

Platforms installed before lifting.
 Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	7278.1 kgm
Shop Test Wt.	- Fabricated weight + water (full)	35379.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	7278.1 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	10118.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	10118.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	27886.5 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	26954.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	38220.1 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	6198867	6095566
20	30	21865484	21676990
30	40	21136636	20954424
40	50	6198867	6095566

Total	55399856.0	54822548.0 mm ²
Total	55.4	54.8 m ²

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-2''	7.94	8.52	85	600	GR 1.4	6.337	5.577	7.943	5.577
N6-2''	1.52	1.63	85	150	GR 1.4	...	...	...	...
N2-3''	3.97	4.26	85	300	GR 1.4	4.878	1.079	3.971	1.079
N7-2''	1.52	1.63	85	150	GR 1.4	...	...	...	...
N4-2''	1.52	1.63	85	150	GR 1.4	...	...	...	...
N3-3''	3.97	4.26	85	300	GR 1.4	4.878	1.079	3.971	1.079
N5-3''	3.97	4.26	85	300	GR 1.4	4.878	1.079	3.971	1.079
M1-24''	1.52	1.63	85	150	GR 1.4	...	...	...	...
N8-3''	1.52	1.63	85	150	GR 1.4	...	...	...	...
M2-24''	1.52	1.63	85	150	GR 1.4	...	...	...	...
Min Rating	1.079	1.630, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1-2''	77.851	0.50	524985	3200	0.707	5.577	Pass
N2-3''	111.125	0.50	1181304	4800	0.707	1.079	Pass
N3-3''	111.125	0.50	1181304	4800	0.707	1.079	Pass
N5-3''	111.125	0.50	1181304	4800	0.707	1.079	Pass

Evaluating Flange on Nozzle [N1-2'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(524985) + 4 \cdot \text{abs}(3200)77.851 \leq \pi \cdot 77.851^3 \cdot [(7.943 - 0.707) + .500 \cdot 7.943]$$

$$= 0.93966E+07 \leq 0.16618E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-2'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 9396616.0 / (\pi \cdot 77.851^3))$$

$$= \min(7.943, 5.577)$$

$$= 5.577 \text{ MPa}$$

Evaluating Flange on Nozzle [N2-3'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 \cdot [(3.971 - 0.707) + .500 \cdot 3.971]$$

$$= 0.21035E+08 \leq 0.22642E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-3'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 21035196.0 / (\pi \cdot 111.125^3))$$

$$= \min(3.971, 1.079)$$

$$= 1.079 \text{ MPa}$$

Evaluating Flange on Nozzle [N3-3'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 \cdot [(3.971 - 0.707) + .500 \cdot 3.971]$$

$$= 0.21035E+08 \leq 0.22642E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-3'' [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 21035196.0 / (\pi \cdot 111.125^3)) \\
 &= \min(3.971, 1.079) \\
 &= 1.079 \text{ MPa}
 \end{aligned}$$

Evaluating Flange on Nozzle [N5-3"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(1181304) + 4 \cdot \text{abs}(4800) 111.125 \leq \pi \cdot 111.125^3 [(3.971 - 0.707) + .500 \cdot 3.971] \\
 &= 0.21035E+08 \leq 0.22642E+08 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N5-3" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 21035196.0 / (\pi \cdot 111.125^3)) \\
 &= \min(3.971, 1.079) \\
 &= 1.079 \text{ MPa}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.84250.75)$
 = 0.111 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.111$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 256139 \cdot 1.5$
 = 25050.381 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 256139$
 = 11133.5 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	1150	42689.8	4175.06
20	30	1150	42689.8	4175.06
20	30	1150	42689.8	4175.06
30	0	1150	42689.8	4175.06
30	40	1150	42689.8	4175.06
40	50	1150	42689.8	4175.06



The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2999.941 mm
Center of Gravity of the Platforms	3000.000 mm
Center of Gravity of the Liquid	2999.991 mm
Center of Gravity of the Insulation	3000.000 mm
Center of Gravity of the Stiffening Rings	3049.966 mm
Center of Gravity of the Nozzles	5500.569 mm
Center of Gravity of the Added Weights (Operating)	3633.955 mm
Center of Gravity of the Added Weights (Empty)	3633.955 mm
Center of Gravity of Bare Shell New and Cold	2999.970 mm
Center of Gravity of Bare Shell Corroded	2999.965 mm
Vessel CG in the Operating Condition	3104.195 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3385.914 mm
Vessel CG in the Test Condition	3079.387 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

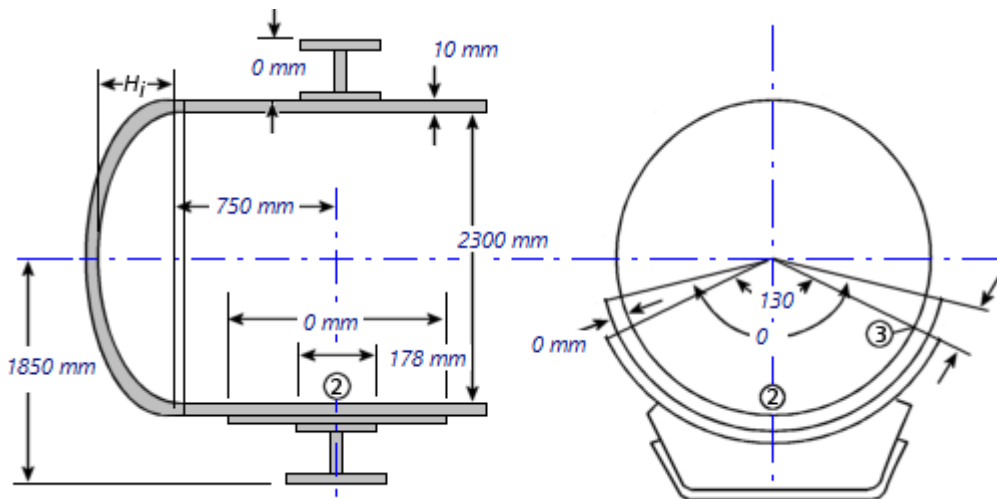
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1156.0 \cdot 8.0}, 2 \cdot 750.0)$$

$$= 328.0197 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1156.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Inside or Mean Depth of Head	Hi or hm	580.00	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	162134.80	N
Pressure used in Analysis	P	0.356 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	22.55	138.00
Long. Stress at Bottom of Midspan	28.85	138.00
Long. Stress at Top of Saddles	29.42	138.00
Long. Stress at Bottom of Saddles	23.61	138.00
Tangential Shear in Shell	11.91	110.40
Circ. Stress at Horn of Saddle	139.96	172.50

Circ. Compressive Stress in Shell	4.48	138.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 = max(Fl,eq; Sum of X Forces) • B / Ls
 = max(25050.38, 0) • 1850.0/4500.0
 = 10298.5 N

Saddle Reaction Force due to Earthquake Ft [Fst]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(25050/2 + 0) • 1850.0/2008.0
 = 34618.9 N

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 127516 + max(0, 0, 10298, 34619)
 = 162134.8 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	166225.44	N
Transverse Shear Load	12525.19	N
Longitudinal Shear Load	25050.38	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2008.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0213	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -162135•750.0[1-(1-750.0/6000.0+(1156.0²-580.0²)/(2•750.0•6000.0))/(1+(4•580.0)/(3•6000.0))]
 = -15386660.0 N-mm

Moment per Equation 4.15.2 [M2]:
 = Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 162135•6000/4((1+2(1156²-580²)/(6000²))/(1+(4•580)/(3•6000))-4•750/6000)
 = 105844656.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:
 = P • Rm/(2t) - M2/(π•Rm²t)
 = 0.356 • 1156.0/(2•8.0) - 0.10584E+09/(π•1156.0²•8.0)
 = 22.55 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 0.356 • 1156.0/(2 • 8.0) + 0.10584E+09/(π • 1156.0² • 8.0)
 = 28.85 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:
 = P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0.356•1156.0/(2•8.0)--15386660.0/(0.1231•π•1156.0²•8.0)
 = 29.42 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0.356•1156.0/(2•8.0)+-15386660.0/(0.2196•π•1156.0²•8.0)
 = 23.61 MPa



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 162135(6000.0 - 2 \cdot 750.0)/(6000.0 + (4 \cdot 580.0/3))$
 $= 107717.5 \text{ N}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.0222 \cdot 107717.5 / (1156.0 \cdot 8.0)$
 $= 11.91 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1156.0 \cdot 8.0)}$
 $= 75.010 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7258 \cdot 162135 \cdot 0.1 / (8.0(178.0 + 75.01 + 75.01))$
 $= -4.48 \text{ MPa}$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:
 $= -Q/(4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$
 $= -162135 / (4 \cdot 8.0(178.0 + 75.01 + 75.01)) -$
 $12 \cdot 0.021 \cdot 162135 \cdot 1156.0 / (6000.0 \cdot 8.0^2)$
 $= -139.96 \text{ MPa}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(178.0 + 1.56 \cdot \sqrt{(1156.0 \cdot 8.0)}, 2 \cdot 750.0)$
 $= 328.02 \text{ mm}$

Distance between Saddle Supports [Ls]:
 $= 4500.0 \text{ mm}$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:
 $= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$
 $= 0.000012 \cdot 4500.0(85.0 - 21.1)$
 $= 3.451 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2008.0001	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	13.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	677.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1152.0 \cdot 8.0)} + 230.0)^3) / 12$
 $= 16203.077 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 16203.077 + 3038.08(316.504 - 4.0)^2$
 $= 296710432.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	379.8	8.0	4.0	3038.1	12152.3	0.297E+09
Wearplate	230.0	10.0	13.0	2300.0	29900.0	0.212E+09
Web	13.0	662.0	349.0	8606.0	3003493.2	0.323E+09

BasePlate	204.0	18.0	689.0	3672.0	2530007.8	0.510E+09
Totals	...	...	...	17616.1	5575553.5	0.134E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5575554 / 17616.078$$

$$= 316.504 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 130.0 / 2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115.0) - 0.5 \cdot \sin(115.0)^2) / (\pi - 2.007 + \sin(115.0) \cos(115.0))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.222 \cdot 162134.797$$

$$= 35964.7422 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.4673 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1982 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1152.0 \cdot \sin(130.0/2) / 1.1345$$

$$= 929.683 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 33449366.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 7.8888 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3313 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(162135 + 4091) \cdot 204.0) / (4 \cdot 2008.0 \cdot 161.331)}$$

$$= 8.861 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130.0/2) (1150.0 + 10.0 + 10.0)$$

$$= 2120.760 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2120.7603 / (3 - 1)$$

$$= 1060.380 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1060.3801 \cdot 204.0 / 2$$

$$= 108158.773 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 166225.422 / (2008.0 \cdot 204.0)$$

$$= 0.406 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 108158.8 \cdot 0.41$$

$$= 43889.977 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 1650.0 + 6892.471$$

$$= 8542.471 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 43890.0/8542.4707$$

$$= 5.138 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	178.0	...	1780.0	...	0.470E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178.0 - 13.0$$

$$= 165.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/8542.471$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4699788/8542.471)}$$

$$= 23.456 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(130.0/2)(1150.0 + 10.0 + 10.0) - 18.0$$

$$= 1337.537 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(130.0/2)(1150.0 + 10.0 + 10.0)$$

$$= 2120.760 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2008.0 - 50.8$$

$$= 1957.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2120.76 - 1957.2) / 2$$

$$= 81.780 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1337.537^2 + 81.78^2}$$

$$= 1340.034 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/242.0)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1340.034/23.456$$

$$= 57.131$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 25050.4 / (2 \cdot 2008.0) \cdot 1060.38 \cdot 1340.03 / 2$$

$$= 4433473.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (57.1305 < 127.7103) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (57.13)^2 / (2 \cdot 127.71^2)) 242 /$$

$$(5/3 + 3 \cdot (57.13) / (8 \cdot 127.71) - (57.13^3) / (8 \cdot 127.71^3))$$

$$= 119.4 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.14 / 119.45 + (4433474 \cdot 89.0 / 4699793) / 161.33$$

$$= 0.563$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	165.0	0.0	1650.0	0.0	0.470E+07
Web	1060.4	13.0	0.0	13784.9	0.0	0.194E+06
Totals	...	...	...	15434.9	...	0.489E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 15434.941$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0 / 2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1850.0 - \cos(1170.0 / 2) (18.0 + 0.0 + 0.0) - 0.0$$

$$= 662.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4892095 / 15434.941) }$$

$$= 17.803 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 662.0 / 17.803$$

$$= 37.185$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 25050.377 / (2 \cdot 2008.0)$$

$$= 6.238 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 6.238 \cdot 1060.38 \cdot 662.0$$

$$= 4380424.500 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4380425 / 54967.414$$

$$= 79.666 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (37.1846 < 127.7103) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.18)^2 / (2 \cdot 127.71^2)) 242 /$$

$$(5/3 + 3 \cdot (37.18) / (8 \cdot 127.71) - (37.18^3) / (8 \cdot 127.71^3))$$

= 130.7 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $5.69/130.72 + (4380425 \cdot 89.0/4892095) / 161.33$
 = 0.537

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	64.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²
Saddle Load QO (Weight)	QO	131606.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	10298.5 N
Maximum Transverse Force	Ft	12525.2 N
Maximum Longitudinal Force	Fl	25050.4 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 86308.1 N

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $\max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $86308.1 / (292.906 \cdot 4)$
 = 73.7 MPa. Must be less than 100.0 MPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $12525.2 / (292.906 \cdot 4)$
 = 10.7 MPa. Must be less than 100.0 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $25050.38 / (100.0 \cdot 4.0)$
 = 62.6313 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1850.0 \cdot 12525.19 + 0.0$
 = 23180996.00 N-mm

Eccentricity (e):
 = $Rmom / QO$
 = $23180996 / 131606.52$
 = 176.07 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 62.6313, 0.0]$
 = 62.6313 mm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: D

Govrn. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(178.0 + 1.56 \cdot \sqrt{1156.0 \cdot 8.0}, 2 \cdot 750.0)$
 $= 328.0197 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1156.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Inside or Mean Depth of Head	Hi or hm	580.00	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		174375.38	N
Pressure used in Analysis	P	0.356	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	22.31	138.00
Long. Stress at Bottom of Midspan	29.09	138.00
Long. Stress at Top of Saddles	29.70	138.00
Long. Stress at Bottom of Saddles	23.45	138.00
Tangential Shear in Shell	12.81	110.40
Circ. Stress at Horn of Saddle	150.52	172.50
Circ. Compressive Stress in Shell	4.82	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(25050.38, 0) \cdot 1850.0 / 4500.0$
 $= 10298.5 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(25050 / 2 + 0) \cdot 1850.0 / 2008.0$
 $= 34618.9 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$

$$= 139756 + \max(0, 0, 10298, 34619)$$

$$= 174375.4 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	178465.81	N
Transverse Shear Load	12525.19	N
Longitudinal Shear Load	25050.38	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2008.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0213	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -174375 \cdot 750.0 [1 - (1 - 750.0/6000.0 + (1156.0^2 - 580.0^2)/(2 \cdot 750.0 \cdot 6000.0))/(1 + (4 \cdot 580.0)/(3 \cdot 6000.0))]$$

$$= -16548295.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 174375 \cdot 6000/4 [(1 + 2(1156^2 - 580^2)/(6000^2))/(1 + (4 \cdot 580)/(3 \cdot 6000)) - 4 \cdot 750/6000]$$

$$= 113835544.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.356 \cdot 1156.0/(2 \cdot 8.0) - 0.11384\text{E}+09/(\pi \cdot 1156.0^2 \cdot 8.0)$$

$$= 22.31 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.356 \cdot 1156.0/(2 \cdot 8.0) + 0.11384\text{E}+09/(\pi \cdot 1156.0^2 \cdot 8.0)$$

$$= 29.09 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.356 \cdot 1156.0/(2 \cdot 8.0) - 16548295.0/(0.1231 \cdot \pi \cdot 1156.0^2 \cdot 8.0)$$

$$= 29.70 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.356 \cdot 1156.0/(2 \cdot 8.0) + 16548295.0/(0.2196 \cdot \pi \cdot 1156.0^2 \cdot 8.0)$$

$$= 23.45 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 174375(6000.0 - 2 \cdot 750.0)/(6000.0 + (4 \cdot 580.0/3))$$

$$= 115849.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.0222 \cdot 115849.78 / (1156.0 \cdot 8.0)$$

$$= 12.81 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1156.0 \cdot 8.0)}$$

$$= 75.010 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$



$$= -0.7258 \cdot 174375 \cdot 0.1 / (8.0(178.0 + 75.01 + 75.01))$$

$$= -4.82 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -174375 / (4 \cdot 8.0(178.0 + 75.01 + 75.01)) -$$

$$12 \cdot 0.021 \cdot 174375 \cdot 1156.0 / (6000.0 \cdot 8.0^2)$$

$$= -150.52 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1156.0 \cdot 8.0}, 2 \cdot 750.0)$$

$$= 328.02 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2008.0001	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	13.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	677.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1152.0 \cdot 8.0} + 230.0)^3) / 12$$

$$= 16203.077 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 16203.077 + 3038.08(316.504 - 4.0)^2$$

$$= 296710432.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	379.8	8.0	4.0	3038.1	12152.3	0.297E+09
Wearplate	230.0	10.0	13.0	2300.0	29900.0	0.212E+09
Web	13.0	662.0	349.0	8606.0	3003493.2	0.323E+09
BasePlate	204.0	18.0	689.0	3672.0	2530007.8	0.510E+09
Totals	...	...	...	17616.1	5575553.5	0.134E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5575554 / 17616.078$$

$$= 316.504 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 130.0 / 2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115.0) - 0.5 \cdot \sin(115.0)^2) / (\pi - 2.007 + \sin(115.0) \cos(115.0))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.222 \cdot 174375.375$$

$$= 38679.9453 \text{ N}$$

Tension Stress, $St = (Fh / A_s) = 2.6535 \text{ MPa}$



Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 145.2000 \text{ MPa}$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1850.0 - 1152.0 \cdot \sin(130.0/2) / 1.1345$
 $= 929.683 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 35974668.0000 \text{ N-mm}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 8.4844 \text{ MPa}$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 161.3333 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 $= \sqrt{\frac{3(174375 + 4090) \cdot 204.0}{4 \cdot 2008.0 \cdot 161.333}}$
 $= 9.181 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 130.0/2) (1150.0 + 10.0 + 10.0)$
 $= 2120.760 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2120.7603 / (3 - 1)$
 $= 1060.380 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 1060.3801 \cdot 204.0 / 2$
 $= 108158.773 \text{ mm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 178465.812 / (2008.0 \cdot 204.0)$
 $= 0.436 \text{ N/mm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 108158.8 \cdot 0.44$
 $= 47121.910 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 1650.0 + 6892.471$
 $= 8542.471 \text{ mm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 47121.9 / 8542.4707$
 $= 5.517 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	178.0	...	1780.0	...	0.470E+07

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 178.0 - 13.0$
 $= 165.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 8542.471$

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 178.0/2

= 89.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{4699788/8542.471}$

= 23.456 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1850.0 - cos(130.0/2)(1150.0 + 10.0 + 10.0) - 18.0

= 1337.537 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(130.0/2)(1150.0 + 10.0 + 10.0)

= 2120.760 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 2008.0 - 50.8

= 1957.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (2120.76 - 1957.2)/2

= 81.780 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1337.537^2 + 81.78^2}$

= 1340.034 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/242.0}$

= 127.710

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1340.034/23.456

= 57.131

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 25050.4 / (2 • 2008.0) • 1060.38 • 1340.03/2

= 4433473.500 N-mm

Compressive Allowable, KL/r < Cc (57.1305 < 127.7095) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (57.13)² / (2 • 127.71²)) 242 /

(5/3 + 3 • (57.13) / (8 • 127.71) - (57.13³) / (8 • 127.71³))

= 119.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 5.52/119.45 + (4433474 • 89.0/4699793) / 161.33

= 0.566

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	165.0	0.0	1650.0	0.0	0.470E+07
Web	1060.4	13.0	0.0	13784.9	0.0	0.194E+06



Totals	...	...	...	15434.9	...	0.489E+07
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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/15434.941$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1170.0/2)(18.0 + 0.0 + 0.0) - 0.0$$

$$= 662.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4892095/15434.941)}$$

$$= 17.803 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 662.0/17.803$$

$$= 37.185$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 25050.377/(2 \cdot 2008.0)$$

$$= 6.238 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 6.238 \cdot 1060.38 \cdot 662.0$$

$$= 4380424.500 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4380425/54967.414$$

$$= 79.666 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (37.1846 < 127.7095) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.18)^2 / (2 \cdot 127.71^2)) 242 /$$

$$(5/3 + 3 \cdot (37.18) / (8 \cdot 127.71) - (37.18^3) / (8 \cdot 127.71^3))$$

$$= 130.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.1/130.72 + (4380425 \cdot 89.0/4892095) / 161.33$$

$$= 0.540$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	64.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²
Saddle Load Q0 (Weight)	Q0	143846.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	10298.5 N
Maximum Transverse Force	Ft	12525.2 N
Maximum Longitudinal Force	Fl	25050.4 N



Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $25050.4 / (292.906 \cdot 4)$
= 21.4 MPa. Must be less than 100.0 MPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $12525.2 / (292.906 \cdot 4)$
= 10.7 MPa. Must be less than 100.0 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $25050.38 / (100.0 \cdot 4.0)$
= 62.6313 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1850.0 \cdot 12525.19 + 0.0$
= 23180996.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $23180996 / 143846.89$
= 161.09 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 62.6313, 0.0]$
= 62.6313 mm²

MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: D

Govern. thk, tg = 10.0, tr = 6.03, c = 2.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.754$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

[Review notes about nozzle loadings and supports in the Warnings report.](#)

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

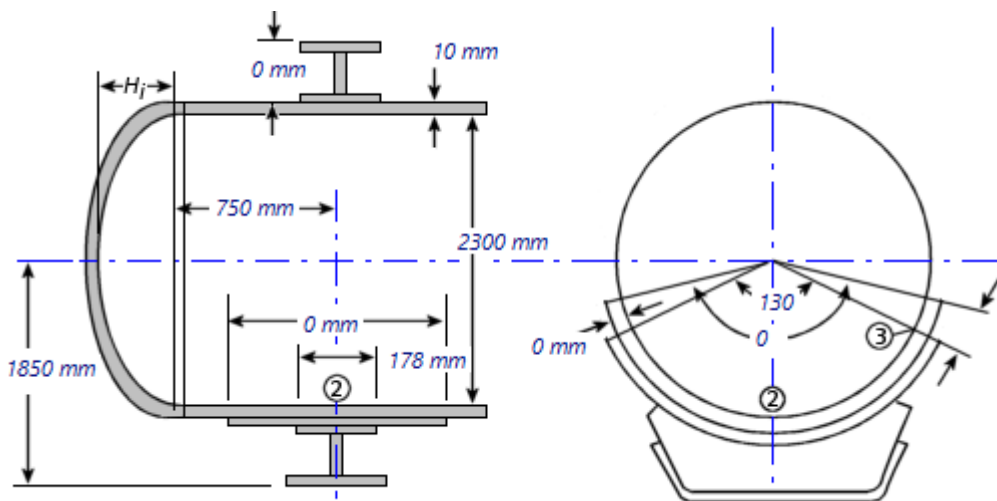
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{1155.0 \cdot 10.0}, 2 \cdot 750.0)$$

$$= 345.6546 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1155.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Inside or Mean Depth of Head	Hi or hm	579.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	177182.17	N
Pressure used in Analysis	P	0.930 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	50.95	234.00
Long. Stress at Bottom of Midspan	56.47	234.00
Long. Stress at Top of Saddles	56.96	234.00
Long. Stress at Bottom of Saddles	51.88	234.00
Tangential Shear in Shell	10.42	138.58
Circ. Stress at Horn of Saddle	99.98	350.99

Circ. Compressive Stress in Shell	3.72	234.00
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There were no external loads for this load condition.

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 177182 + max(0, 0, 0, 0)
 = 177182.2 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	181272.81	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	0.00	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 2008.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0213	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-177182 \cdot 750.0 [1 - (1 - 750.0/6000.0 + (1155.0^2 - 579.0^2)/(2 \cdot 750.0 \cdot 6000.0))/(1 + (4 \cdot 579.0)/(3 \cdot 6000.0))]$
 = -16806880.0 N-mm

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $177182 \cdot 6000/4 ((1 + 2(1155^2 - 579^2)/(6000^2))/(1 + (4 \cdot 579)/(3 \cdot 6000)) - 4 \cdot 750/6000)$
 = 115701736.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $0.93 \cdot 1155.0/(2 \cdot 10.0) - 0.11570E+09/(\pi \cdot 1155.0^2 \cdot 10.0)$
 = 50.95 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.93 \cdot 1155.0/(2 \cdot 10.0) + 0.11570E+09/(\pi \cdot 1155.0^2 \cdot 10.0)$
 = 56.47 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 = $0.93 \cdot 1155.0/(2 \cdot 10.0) - 16806880.0/(0.1231 \cdot \pi \cdot 1155.0^2 \cdot 10.0)$
 = 56.96 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= $P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$
 = $0.93 \cdot 1155.0/(2 \cdot 10.0) + 16806880.0/(0.2196 \cdot \pi \cdot 1155.0^2 \cdot 10.0)$
 = 51.88 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$
 = $177182(6000.0 - 2 \cdot 750.0)/(6000.0 + (4 \cdot 579.0/3))$
 = 117737.7 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (Rm \cdot t)$
 = $1.0222 \cdot 117737.71/(1155.0 \cdot 10.0)$
 = 10.42 MPa

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1155.0 \cdot 10.0)}$$

$$= 83.827 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7258 \cdot 177182 \cdot 0.1 / (10.0(178.0 + 83.83 + 83.83))$$

$$= -3.72 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -177182 / (4 \cdot 10.0(178.0 + 83.827 + 83.827)) -$$

$$12 \cdot 0.021 \cdot 177182 \cdot 1155.0 / (6000.0 \cdot 10.0^2)$$

$$= -99.98 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{(1155.0 \cdot 10.0)}, 2 \cdot 750.0)$$

$$= 345.65 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2008.0001	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	13.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	677.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1150.0 \cdot 10.0)} + 230.0)10.0^3) / 12$$

$$= 33107.574 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 33107.574 + 3972.913(302.542 - 5.0)^2$$

$$= 351758752.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	397.3	10.0	5.0	3972.9	19864.6	0.352E+09
Wearplate	230.0	10.0	15.0	2300.0	34500.0	0.190E+09
Web	13.0	662.0	351.0	8606.0	3020706.0	0.335E+09
BasePlate	204.0	18.0	691.0	3672.0	2537352.0	0.554E+09
Totals	...	...	...	18550.9	5612423.0	0.143E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5612423 / 18550.914$$

$$= 302.542 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 130.0 / 2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(115.0) - 0.5 \cdot \sin(115.0)^2) / (\pi - 2.007 + \sin(115.0)\cos(115.0))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.222 \cdot 177182.172$$

$$= 39302.5508 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.6962 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$$

$$= 1850.0 - 1150.0 \cdot \sin(130.0/2)/1.1345$$

$$= 931.281 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 36616548.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 7.7409 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(177182 + 4091) \cdot 204.0) / (4 \cdot 2008.0 \cdot 173.33)}$$

$$= 8.927 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130.0/2) (1150.0 + 10.0 + 10.0)$$

$$= 2120.760 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2120.7603 / (3 - 1)$$

$$= 1060.380 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1060.3801 \cdot 204.0 / 2$$

$$= 108158.773 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 181272.812 / (2008.0 \cdot 204.0)$$

$$= 0.443 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 108158.8 \cdot 0.44$$

$$= 47863.070 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 1650.0 + 6892.471$$

$$= 8542.471 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 47863.1/8542.4707$$

$$= 5.603 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	178.0	...	1780.0	...	0.470E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178.0 - 13.0$$

$$= 165.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0/8542.471 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 178.0/2 \\ &= 89.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(4699788/8542.471)} \\ &= 23.456 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 1850.0 - \cos(130.0/2)(1150.0 + 10.0 + 10.0) - 18.0 \\ &= 1337.537 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\ &= 2 \cdot \sin(130.0/2)(1150.0 + 10.0 + 10.0) \\ &= 2120.760 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= \text{Baseplate length} - \text{clearance} \\ &= 2008.0 - 50.8 \\ &= 1957.200 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (Lw - Bd) / 2 \\ &= (2120.76 - 1957.2) / 2 \\ &= 81.780 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{Lv^2 + Dd^2} \\ &= \sqrt{1337.537^2 + 81.78^2} \\ &= 1340.034 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 199955/260.0)} \\ &= 123.211 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 1340.034/23.456 \\ &= 57.131 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\ &= 0.0 / (2 \cdot 2008.0) \cdot 1060.38 \cdot 1340.03/2 \\ &= 0.000 \text{ N-mm} \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($57.1305 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\ &= (1 - (57.13)^2 / (2 \cdot 123.21^2)) 260 / \\ &\quad (5/3 + 3 \cdot (57.13) / (8 \cdot 123.21) - (57.13^3) / (8 \cdot 123.21^3)) \\ &= 126.9 \text{ MPa} \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\ &= 5.6/126.93 + (0.0 \cdot 89.0/4699793) / 173.33 \\ &= 0.044 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	165.0	0.0	1650.0	0.0	0.470E+07
Web	1060.4	13.0	0.0	13784.9	0.0	0.194E+06
Totals	...	...	...	15434.9	...	0.489E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/15434.941$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1170.0/2)(18.0 + 0.0 + 0.0) - 0.0$$

$$= 662.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4892095/15434.941)}$$

$$= 17.803 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 662.0/17.803$$

$$= 37.185$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 0.0/(2 \cdot 2008.0)$$

$$= 0.000 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.0 \cdot 1060.38 \cdot 662.0$$

$$= 0.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0.0/54967.414$$

$$= 0.000 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($37.1846 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.18)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (37.18) / (8 \cdot 123.21) - (37.18^3) / (8 \cdot 123.21^3))$$

$$= 139.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 6.2/139.7 + (0.0 \cdot 89.0/4892095) / 173.33$$

$$= 0.044$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	64.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²



Saddle Load Q0 (Weight)	Q0	181272.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	0.0 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	0.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $0.0 / (292.906 \cdot 4)$
 = 0.0 MPa. Must be less than 68.4 MPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $0.0 / (292.906 \cdot 4)$
 = 0.0 MPa. Must be less than 68.4 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $0.0 / (68.4 \cdot 4.0)$
 = 0.0000 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1850.0 \cdot 0.0 + 0.0$
 = 0.00 N-mm

Eccentricity (e):
 = $Rmom / Q0$
 = $0.0 / 181272.81$
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[0.0, 0.0, 0.0]$
 = 0.0000 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 = $\min(178.0 + 1.56 \cdot \sqrt{ 1155.0 \cdot 10.0 }, 2 \cdot 750.0)$
 = 345.6546 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1155.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Inside or Mean Depth of Head	Hi or hm	579.00	mm



Shell Allowable Stress used in Calculation 234.00 MPa
 Head Allowable Stress used in Calculation 234.00 MPa
 Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1850.00 mm
 Coefficient of Friction mu 0.60

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 189423.38 N
 Pressure used in Analysis P 0.930 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	50.76	234.00
Long. Stress at Bottom of Midspan	56.66	234.00
Long. Stress at Top of Saddles	57.19	234.00
Long. Stress at Bottom of Saddles	51.75	234.00
Tangential Shear in Shell	11.14	138.58
Circ. Stress at Horn of Saddle	106.88	350.99
Circ. Compressive Stress in Shell	3.98	234.00

There were no external loads for this load condition.

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(FwL, Fwt, Fsl, Fst)
 = 189423 + max(0, 0, 0, 0)
 = 189423.4 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 193513.81 N
 Transverse Shear Load 0.00 N
 Longitudinal Shear Load 0.00 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2008.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231 K2 = 1.0222 K3 = 0.7217 K4 = 0.3619
 K5 = 0.7258 K6 = 0.0449 K7 = 0.0213 K8 = 0.3296
 K9 = 0.2551 K10 = 0.0508 K1* = 0.2196

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/(3L))]$$

$$= -189423 \cdot 750.0 [1 - (1 - 750.0/6000.0 + (1155.0^2 - 579.0^2)/(2 \cdot 750.0 \cdot 6000.0)) / (1 + (4 \cdot 579.0)/(3 \cdot 6000.0))]$$

$$= -17968040.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \{ (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L \}$$

$$= 189423 \cdot 6000 / 4 \{ (1 + 2(1155^2 - 579^2)/(6000^2)) / (1 + (4 \cdot 579)/(3 \cdot 6000)) - 4 \cdot 750 / 6000 \}$$

$$= 123695384.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 0.93 \cdot 1155.0 / (2 \cdot 10.0) - 0.12370E+09 / (\pi \cdot 1155.0^2 \cdot 10.0)$$

$$= 50.76 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$



$$= 0.93 \cdot 1155.0 / (2 \cdot 10.0) + 0.12370E+09 / (\pi \cdot 1155.0^2 \cdot 10.0)$$

$$= 56.66 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.93 \cdot 1155.0 / (2 \cdot 10.0) - 17968040.0 / (0.1231 \cdot \pi \cdot 1155.0^2 \cdot 10.0)$$

$$= 57.19 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.93 \cdot 1155.0 / (2 \cdot 10.0) + 17968040.0 / (0.2196 \cdot \pi \cdot 1155.0^2 \cdot 10.0)$$

$$= 51.75 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 189423(6000.0 - 2 \cdot 750.0) / (6000.0 + (4 \cdot 579.0 / 3))$$

$$= 125872.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.0222 \cdot 125872.01 / (1155.0 \cdot 10.0)$$

$$= 11.14 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1155.0 \cdot 10.0)}$$

$$= 83.827 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7258 \cdot 189423 \cdot 0.1 / (10.0(178.0 + 83.83 + 83.83))$$

$$= -3.98 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -189423 / (4 \cdot 10.0(178.0 + 83.827 + 83.827)) - 12 \cdot 0.021 \cdot 189423 \cdot 1155.0 / (6000.0 \cdot 10.0^2)$$

$$= -106.88 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(178.0 + 1.56 \cdot \sqrt{(1155.0 \cdot 10.0)}, 2 \cdot 750.0)$$

$$= 345.65 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2008.0001	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	13.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	677.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1150.0 \cdot 10.0)} + 230.0)^3) / 12$$

$$= 33107.574 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 33107.574 + 3972.913(302.542 - 5.0)^2$$

$$= 351758752.000 \text{ mm}^4$$

B

D

Y

A

AY

I + AD^(2)

	mm	mm	mm	mm ²	mm ³	mm ⁴
Shell	397.3	10.0	5.0	3972.9	19864.6	0.352E+09
Wearplate	230.0	10.0	15.0	2300.0	34500.0	0.190E+09
Web	13.0	662.0	351.0	8606.0	3020706.0	0.335E+09
BasePlate	204.0	18.0	691.0	3672.0	2537352.0	0.554E+09
Totals	...	...	...	18550.9	5612423.0	0.143E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5612423/18550.914$$

$$= 302.542 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 130.0/2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115.0) - 0.5 \cdot \sin(115.0)^2) / (\pi - 2.007 + \sin(115.0) \cos(115.0))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.222 \cdot 189423.375$$

$$= 42017.8984 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.8825 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1150.0 \cdot \sin(130.0/2) / 1.1345$$

$$= 931.281 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 39146324.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8.2757 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(189423 + 4090)204.0 / (4 \cdot 2008.0 \cdot 173.33)}$$

$$= 9.224 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130.0/2) (1150.0 + 10.0 + 10.0)$$

$$= 2120.760 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2120.7603 / (3 - 1)$$

$$= 1060.380 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1060.3801 \cdot 204.0/2$$

$$= 108158.773 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 193513.797 / (2008.0 \cdot 204.0)$$

$$= 0.472 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 108158.8 \cdot 0.47$$

$$= 51095.164 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 1650.0 + 6892.471$$

$$= 8542.471 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 51095.2/8542.4707$$

$$= 5.982 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	178.0	...	1780.0	...	0.470E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178.0 - 13.0$$

$$= 165.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/8542.471$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4699788/8542.471)}$$

$$= 23.456 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(130.0/2)(1150.0 + 10.0 + 10.0) - 18.0$$

$$= 1337.537 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(130.0/2)(1150.0 + 10.0 + 10.0)$$

$$= 2120.760 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2008.0 - 50.8$$

$$= 1957.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2120.76 - 1957.2) / 2$$

$$= 81.780 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1337.537^2 + 81.78^2}$$

$$= 1340.034 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1340.034/23.456$$

$$= 57.131$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 0.0 / (2 \cdot 2008.0) \cdot 1060.38 \cdot 1340.03/2$$

$$= 0.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (57.1305 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (57.13)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (57.13) / (8 \cdot 123.21) - (57.13^3) / (8 \cdot 123.21^3))$$

$$= 126.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.98/126.93 + (0.0 \cdot 89.0/4699793) / 173.33$$

$$= 0.047$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	165.0	0.0	1650.0	0.0	0.470E+07
Web	1060.4	13.0	0.0	13784.9	0.0	0.194E+06
Totals	...	...	...	15434.9	...	0.489E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/15434.941$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178.0/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1850.0 - \cos(1170.0/2) (18.0 + 0.0 + 0.0) - 0.0$$

$$= 662.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4892095/15434.941) }$$

$$= 17.803 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 662.0/17.803$$

$$= 37.185$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 0.0 / (2 \cdot 2008.0)$$

$$= 0.000 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.0 \cdot 1060.38 \cdot 662.0$$

$$= 0.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0.0/54967.414$$

$$= 0.000 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (37.1846 < 123.2107) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (37.18)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (37.18) / (8 \cdot 123.21) - (37.18^3) / (8 \cdot 123.21^3))$
 $= 139.7 \text{ MPa}$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 $= 6.62 / 139.7 + (0.0 \cdot 89.0 / 4892095) / 173.33$
 $= 0.047$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	64.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²
Saddle Load QO (Weight)	QO	193513.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	0.0 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	Fl	0.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0.0 / (292.906 \cdot 4)$
 $= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0.0 / (292.906 \cdot 4)$
 $= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 0.0 / (68.4 \cdot 4.0)$
 $= 0.0000 \text{ mm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1850.0 \cdot 0.0 + 0.0$
 $= 0.00 \text{ N-mm}$

Eccentricity (e):
 $= Rmom / QO$
 $= 0.0 / 193513.81$
 $= 0.00 \text{ mm} < B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0.0, 0.0, 0.0]$
 $= 0.0000 \text{ mm}^2$



Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1-2''

From: 10

Pressure for Reinforcement Calculations	P	0.7083	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	2.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	600.0000	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

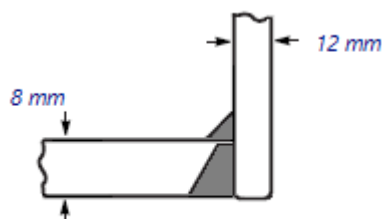
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600
Flange Grade		GR 1.4

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N1-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.71 \cdot 2304.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.71)$
 $= 5.3156 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 27.4 / (118 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 0.1651 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3567 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	114.6034 mm
Parallel to Vessel Wall, opening length	d	57.3017 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, tg = 8.0, tr = 5.316, c = 2.0 mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.886$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: N1-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.71 \cdot 2304.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.71)$
 $= 5.3156 \text{ mm}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 27.4 / (118 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 0.1651 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3567 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.6000	mm
Parallel to Vessel Wall, opening length	d	54.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N1-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3567	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.9024	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.9024	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.8329	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 6.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9024, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.3567, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	277.8 MPa	Passed
Occasional	:	0.8,	Allowable	:	156.9 MPa	Passed
Shear	:	14.8,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N1-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.6749 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.0300 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N1-2'' :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	2.0000	mm
Vessel Diameter	Dv	4140.000	mm
Vessel Thickness	Tv	8.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	2.0000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	12.000	mm
Nozzle Material		SA-333 6	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

Design Internal Pressure	Dp	0.708	MPa
Include Pressure Thrust		No	

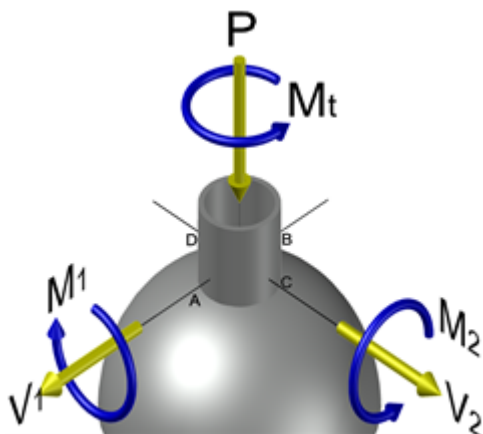
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3199.9	N
Longitudinal Shear (SUS)	(V1) V1	2828.3	N
Circumferential Shear (SUS)	(Vc) V2	2828.3	N
Circumferential Moment (SUS)	(Mc) M1	371220.7	N-mm
Longitudinal Moment (SUS)	(Ml) M2	371220.7	N-mm
Torsional Moment (SUS)	Mt	480169.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 74.8 + 2 \cdot 1.65 \cdot \sqrt{(2075.0 (8.0 - 2.0))}$
 $= 443.013 \text{ mm}$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-3199.9	N
Circumferential Shear	(VC) V2	2828.3	N
Longitudinal Shear	(VL) V1	2828.3	N
Circumferential Moment	(MC) M1	371220.7	N-mm
Longitudinal Moment	(ML) M2	371220.7	N-mm
Torsional Moment	MT	480169.5	N-mm

Dimensionless Param: U = 0.34 TAU = 5.00 (3.24) RHO = 0.60

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.09763	(A,B,C,D)
M(x) / P	SP 1	0.12860	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.15257	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.40361	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.15257	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.40361	(A,B,C,D)
N(y) * T / P	SP 1	0.15248	(A,B,C,D)
M(y) / P	SP 1	0.04327	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.07341	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.14890	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.07341	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.14890	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	8.7	8.7	8.7	8.7	8.7	8.7	8.7	8.7
Rad.	Bend. P	68.6	-68.6	68.6	-68.6	68.6	-68.6	68.6	-68.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-14.1	-14.1	14.1	14.1
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-223.7	223.7	223.7	-223.7
Rad.	Memb. ML	-14.1	-14.1	14.1	14.1	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-223.7	223.7	223.7	-223.7	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-160.6	149.7	315.1	-269.5	-160.6	149.7	315.1	-269.5
Tang.	Memb. P	13.6	13.6	13.6	13.6	13.6	13.6	13.6	13.6
Tang.	Bend. P	23.1	-23.1	23.1	-23.1	23.1	-23.1	23.1	-23.1
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-6.8	-6.8	6.8	6.8
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-82.5	82.5	82.5	-82.5
Tang.	Memb. ML	-6.8	-6.8	6.8	6.8	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-82.5	82.5	82.5	-82.5	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-52.7	66.2	126.0	-85.3	-52.7	66.2	126.0	-85.3
	Shear VC	4.0	4.0	-4.0	-4.0	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-4.0	-4.0	4.0	4.0
	Shear MT	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
	Tot. Shear	13.1	13.1	5.1	5.1	5.1	5.1	13.1	13.1
	Str. Int.	162.1	151.7	315.2	269.7	160.8	150.0	316.0	270.5



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Rad.	Pl (SUS)	-5.4	-5.4	22.8	22.8	-5.4	-5.4	22.8	22.8
Rad.	Q (SUS)	-155.1	155.1	292.3	-292.3	-155.1	155.1	292.3	-292.3
Long.	Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Long.	Pl (SUS)	6.8	6.8	20.3	20.3	6.8	6.8	20.3	20.3
Long.	Q (SUS)	-59.5	59.5	105.6	-105.6	-59.5	59.5	105.6	-105.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	4.0	4.0	-4.0	-4.0	-4.0	-4.0	4.0	4.0
Shear	Q (SUS)	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
	Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
	Pm+Pl (SUS)	130.3	130.3	148.1	148.1	130.3	130.3	148.1	148.1
	Pm+Pl+Q (Total)	111.0	274.0	437.5	184.5	108.4	272.3	438.3	186.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.30	138.00	Passed
Pm+Pl (SUS)	148.05	207.00	Passed
Pm+Pl+Q (TOTAL)	438.30	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N6-2''

From: 10

Pressure for Reinforcement Calculations	P	0.7174	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	2.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	750.0000	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

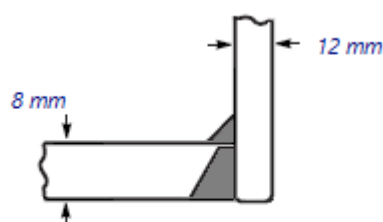
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150
Flange Grade		GR 1.4

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N6-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.72 \cdot 2304.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.72)$
 $= 5.3834 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.72 \cdot 27.4 / (118 \cdot 1.0 - 0.6 \cdot 0.72)$
 $= 0.1672 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3567 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	119.2862 mm
Parallel to Vessel Wall, opening length	d	59.6431 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, tg = 8.0, tr = 5.383, c = 2.0 mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.897$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0000 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.72 \cdot 2304.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.72)$
 $= 5.3834 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.72 \cdot 27.4 / (118 \cdot 1.0 - 0.6 \cdot 0.72)$
 $= 0.1672 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3567 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.6000	mm
Parallel to Vessel Wall, opening length	d	54.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3567	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.9776	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.9776	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.8329	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 6.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9776, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.3567, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell welds	Tmin	8.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot \text{tmin.}$	$7.0700 = 0.7 \cdot \text{Wo mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 14.9055 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.4825 mm

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Input, Nozzle Desc: N2-3''

From: 20

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	1191.6666	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

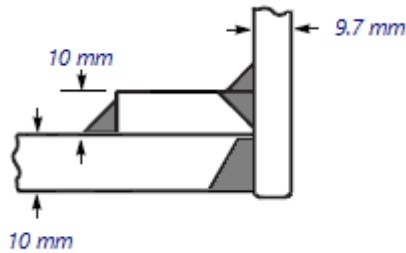
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	150.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		30.5500	mm

Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2-3''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9173 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.2656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6286 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.8618 mm
Parallel to Vessel Wall, opening length	d	73.4309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: N2-3''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6286 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9173 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9173 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8352 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 6.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9173, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.6286, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	31.1,	Allowable :	118.0 MPa	Passed
Expansion :	0.0,	Allowable :	263.9 MPa	Passed
Occasional :	1.5,	Allowable :	156.9 MPa	Passed
Shear :	23.2,	Allowable :	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
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Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $0.71/4.26 = 0.166$

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7345	mm
Intermediate Calc. for pad/shell Welds	TminPad	8.0000	mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$5.4142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$4.0000 = 0.5 * t_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.8593 mm

Input Echo, WRC297 Item 1, Description: N2-3'' :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	88.900	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	150.000	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.71	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.132) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1161.0(10.0 - 2.0))}$
 $= 406.935$ mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.000	mm
Nozzle Outside Diameter	(d) =	88.900	mm



Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 7.735 mm
 T / t = 2.327
 d / t = 11.494

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.435$$

Nr/P = 0.203
 Mr/P = 0.195
 M0/P = 0.140
 N0/P = 0.185
 MrD/Mc = 0.216
 NrDL/Mc = 0.189
 M0d/Mc = 0.189
 N0DL/Mc = -0.112
 MrD/Ml = 0.204
 NrDL/Ml = 0.184
 M0D/Ml = 0.177
 N0DL/Ml = -0.051

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	44	45	44	45
Outplane Stress Summary	27	72	92	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	-2	41	88	-25

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	32	73	94	26

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	12	43	63	-18

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-32	32	32	-32
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	45	44	45
Inplane Stress Summary	28	58	102	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	59	103	25

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	8	8	4	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-118	118	118	-118
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-94	51	192	-133

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	106	56	193	140

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	9	10	3	4

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-98	98	98	-98
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-68	38	167	-120

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11
Shear Stress Summary	6	6	15	15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	79	39	169	128

Stress Computations at the Edge of the Reinforcing Pad:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1156.0(10.0 - 2.0))}$$

$$= 406.249 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2312.000 mm
Nozzle Outside Diameter	(d) =	150.000 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	38.285 mm
T / t	=	0.209
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.103$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.102
Mr/P	=	0.215
M0/P	=	0.062
N0/P	=	0.111
MrD/Mc	=	0.310
NrDL/Mc	=	0.073
M0d/Mc	=	0.095
N0DL/Mc	=	0.050
MrD/Ml	=	0.181
NrDL/Ml	=	0.097
M0D/Ml	=	0.056
N0DL/Ml	=	0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	101	102	101	102
Outplane Stress Summary	94	104	181	60

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	50	50	50	50
Inplane Stress Summary	39	58	270	-134

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3



Shear Stress Summary 5 5 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 94 105 270 195

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50

Outplane Stress Summary 40 69 133 -7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-139	139	139	-139
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	101	102	101	102

Inplane Stress Summary 60 146 350 -120

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 1 1 6 6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 60 146 350 121

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	44	45	44	45	44	45	44	45
Circ. Pl (SUS)	3	3	1	1	-2	-2	6	6
Circ. Q (SUS)	-22	22	46	-46	-15	15	49	-49



Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-3	-3	7	7	4	4	0	0
Long. Q (SUS)	-22	22	56	-56	-16	16	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	44.0	45.0	44.0	45.0	44.0	45.0	44.0	45.0
Pm+Pl (SUS)	47.0	48.0	45.1	46.1	42.1	43.1	50.0	51.0
Pm+Pl+Q (Total)	29.7	70.8	92.2	27.7	27.5	58.5	99.7	22.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	138.00	Passed
Pm+Pl (SUS)	51.03	207.00	Passed
Pm+Pl+Q (TOTAL)	99.66	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	101	102	101	102	101	102	101	102
Circ. Pl (SUS)	-2	-2	18	18	2	2	12	12
Circ. Q (SUS)	-4	4	60	-60	-43	43	235	-235
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-2	-2	16	16	5	5	11	11
Long. Q (SUS)	-9	9	201	-201	-14	14	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	101.0	102.0	101.0	102.0	101.0	102.0	101.0	102.0
Pm+Pl (SUS)	99.0	100.0	119.0	120.0	103.1	104.1	113.1	114.1
Pm+Pl+Q (Total)	95.3	104.3	267.0	195.0	60.1	147.0	348.1	121.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	120.02	207.00	Passed
Pm+Pl+Q (TOTAL)	348.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	3	3	1	1	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-23	-23	27	27	-17	-17	21	21
Long. Q (SUS)	-73	73	163	-163	-53	53	143	-143
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	28.6	29.6	28.4	28.4	24.4	25.3	22.8	22.8
Pm+Pl+Q (Total)	104.8	55.1	191.3	140.9	77.3	38.6	166.4	128.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	29.61	177.00	Passed
Pm+Pl+Q (TOTAL)	191.34	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N7-2''

From: 20

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	2049.9001	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1250.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

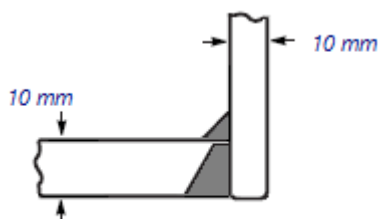
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150
Flange Grade		GR 1.4

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N7-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9173 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.1517 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4306 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	69.6000	mm
Parallel to Vessel Wall, opening length	d	34.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N7-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4306	mm
Wall Thickness per UG16(b),	tr16b = 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9173	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9173	mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8352	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm
Wall Thickness per table UG-45	tb3 = 5.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.42, \max(7.9173, 3.5)]$
 $= 5.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4306, 5.42)$
 $= 5.4200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: N7-2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$



Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2805 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.2805 mm

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Input, Nozzle Desc: N4-2''

From: 20

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	2049.9001	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2850.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

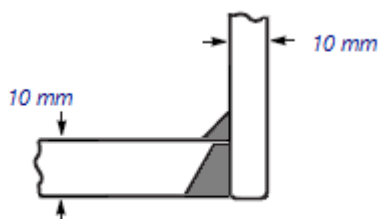
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150
Flange Grade		GR 1.4

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9173 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.1517 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4306 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	69.6000	mm
Parallel to Vessel Wall, opening length	d	34.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N4-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4306	mm
Wall Thickness per UG16(b),	tr16b = 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9173	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9173	mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8352	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm
Wall Thickness per table UG-45	tb3 = 5.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.42, \max(7.9173, 3.5)]$
 $= 5.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.4306, 5.42)$
 $= 5.4200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: N4-2''

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$



Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2805 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.2805 mm

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Input, Nozzle Desc: N3-3"
From: 20

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	2049.9001	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1850.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

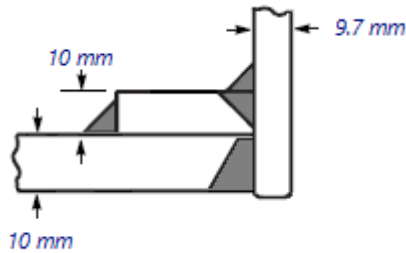
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	150.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		30.5500	mm

Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-3''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9173 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.2656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6286 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.8618 mm
Parallel to Vessel Wall, opening length	d	73.4309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: N3-3''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6286 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9173 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9173 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8352 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 6.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9173, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.6286, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	31.1,	Allowable :	118.0 MPa	Passed
Expansion :	0.0,	Allowable :	263.9 MPa	Passed
Occasional :	1.5,	Allowable :	156.9 MPa	Passed
Shear :	23.2,	Allowable :	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $0.71/4.26 = 0.166$

Weld Size Calculations, Description: N3-3"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7345	mm
Intermediate Calc. for pad/shell Welds	TminPad	8.0000	mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$4.0000 = 0.5 * t_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.8593 mm

Input Echo, WRC297 Item 1, Description: N3-3'' :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	88.900	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	150.000	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	0.71	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (9.132) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 * 1.65 * \sqrt{(R_{mean}(t - ca))}$
 $= 88.9 + 2 * 1.65 * \sqrt{(1161.0(10.0 - 2.0))}$
 $= 406.935$ mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2322.000 mm



Nozzle Outside Diameter (d) = 88.900 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 7.735 mm
 T / t = 2.327
 d / t = 11.494

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.435$$

Nr/P = 0.203
 Mr/P = 0.195
 M0/P = 0.140
 N0/P = 0.185
 MrD/Mc = 0.216
 NrDL/Mc = 0.189
 M0d/Mc = 0.189
 N0DL/Mc = -0.112
 MrD/Ml = 0.204
 NrDL/Ml = 0.184
 M0D/Ml = 0.177
 N0DL/Ml = -0.051

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	44	45	44	45
Outplane Stress Summary	27	72	92	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	-2	41	88	-25

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	32	73	94	26

Vessel Stresses (MPa):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	12	43	63	-18

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-32	32	32	-32
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	45	44	45
Inplane Stress Summary	28	58	102	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	59	103	25

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	8	8	4	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-118	118	118	-118
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-94	51	192	-133
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	14	14	7	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	106	56	193	140
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	9	10	3	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-98	98	98	-98
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-68	38	167	-120
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11

Shear Stress Summary	6	6	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	79	39	169	128
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{1156.0(10.0 - 2.0)}$$

$$= 406.249 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2312.000 mm
Nozzle Outside Diameter	(d) =	150.000 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	38.285 mm
T / t	=	0.209
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.103$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.102
Mr/P	=	0.215
M0/P	=	0.062
N0/P	=	0.111
MrD/Mc	=	0.310
NrDL/Mc	=	0.073
M0d/Mc	=	0.095
N0DL/Mc	=	0.050
MrD/M1	=	0.181
NrDL/M1	=	0.097
M0D/M1	=	0.056
N0DL/M1	=	0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	101	102	101	102
Outplane Stress Summary	94	104	181	60

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	50	50	50	50
Inplane Stress Summary	39	58	270	-134

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	94	105	270	195

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50
Outplane Stress Summary	40	69	133	-7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-139	139	139	-139
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	101	102	101	102
Inplane Stress Summary	60	146	350	-120

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	60	146	350	121

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	44	45	44	45	44	45	44	45
Circ. Pl (SUS)	3	3	1	1	-2	-2	6	6



Circ. Q (SUS)	-22	22	46	-46	-15	15	49	-49
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-3	-3	7	7	4	4	0	0
Long. Q (SUS)	-22	22	56	-56	-16	16	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	44.0	45.0	44.0	45.0	44.0	45.0	44.0	45.0
Pm+Pl (SUS)	47.0	48.0	45.1	46.1	42.1	43.1	50.0	51.0
Pm+Pl+Q (Total)	29.7	70.8	92.2	27.7	27.5	58.5	99.7	22.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	138.00	Passed
Pm+Pl (SUS)	51.03	207.00	Passed
Pm+Pl+Q (TOTAL)	99.66	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	101	102	101	102	101	102	101	102
Circ. Pl (SUS)	-2	-2	18	18	2	2	12	12
Circ. Q (SUS)	-4	4	60	-60	-43	43	235	-235
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-2	-2	16	16	5	5	11	11
Long. Q (SUS)	-9	9	201	-201	-14	14	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	101.0	102.0	101.0	102.0	101.0	102.0	101.0	102.0
Pm+Pl (SUS)	99.0	100.0	119.0	120.0	103.1	104.1	113.1	114.1
Pm+Pl+Q (Total)	95.3	104.3	267.0	195.0	60.1	147.0	348.1	121.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	120.02	207.00	Passed
Pm+Pl+Q (TOTAL)	348.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	3	3	1	1	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-23	-23	27	27	-17	-17	21	21
Long. Q (SUS)	-73	73	163	-163	-53	53	143	-143
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	28.6	29.6	28.4	28.4	24.4	25.3	22.8	22.8
Pm+Pl+Q (Total)	104.8	55.1	191.3	140.9	77.3	38.6	166.4	128.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	29.61	177.00	Passed
Pm+Pl+Q (TOTAL)	191.34	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-3''

From: 30

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	2050.1003	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4250.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

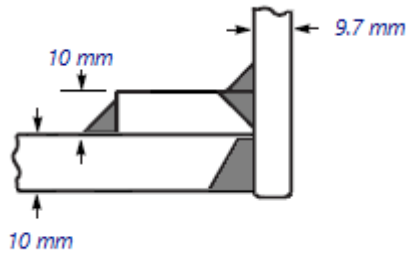
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	150.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		30.5500	mm

Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N5-3''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9172 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.2656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6286 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.8618 mm
Parallel to Vessel Wall, opening length	d	73.4309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: N5-3''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6286 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9172 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9172 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8351 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 6.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9172, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.6286, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	31.1,	Allowable :	118.0 MPa	Passed
Expansion :	0.0,	Allowable :	263.9 MPa	Passed
Occasional :	1.5,	Allowable :	156.9 MPa	Passed
Shear :	23.2,	Allowable :	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.735$, $t_r = 0.266$, $c = 2.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 10.0$, $t_r = 5.917$, $c = 2.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $t_g = 9.735$, $t_r = 0.266$, $c = 2.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $0.71/4.26 = 0.166$

Weld Size Calculations, Description: N5-3"

Intermediate Calc. for nozzle/shell Welds	T_{min}	7.7345	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	8.0000	mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$4.0000 = 0.5 * t_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.8593 mm

Input Echo, WRC297 Item 1, Description: N5-3'' :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	2.0000	mm
Nozzle Diameter	88.900	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	150.000	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	0.71	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (9.132) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1161.0(10.0 - 2.0))}$
 $= 406.935$ mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2322.000 mm



Nozzle Outside Diameter (d) = 88.900 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 7.735 mm
 T / t = 2.327
 d / t = 11.494

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.435$$

Nr/P = 0.203
 Mr/P = 0.195
 M0/P = 0.140
 N0/P = 0.185
 MrD/Mc = 0.216
 NrDL/Mc = 0.189
 M0d/Mc = 0.189
 N0DL/Mc = -0.112
 MrD/Ml = 0.204
 NrDL/Ml = 0.184
 M0D/Ml = 0.177
 N0DL/Ml = -0.051

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	44	45	44	45
Outplane Stress Summary	27	72	92	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	-2	41	88	-25

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	32	73	94	26

Vessel Stresses (MPa):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	12	43	63	-18

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-32	32	32	-32
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	45	44	45
Inplane Stress Summary	28	58	102	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	59	103	25

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	8	8	4	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-118	118	118	-118
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-94	51	192	-133
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	14	14	7	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	106	56	193	140
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	9	10	3	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-98	98	98	-98
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-68	38	167	-120
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11

Shear Stress Summary	6	6	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	79	39	169	128
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{1156.0(10.0 - 2.0)}$$

$$= 406.249 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2312.000 mm
Nozzle Outside Diameter	(d) =	150.000 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	38.285 mm
T / t	=	0.209
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.103$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.102
Mr/P	=	0.215
M0/P	=	0.062
N0/P	=	0.111
MrD/Mc	=	0.310
NrDL/Mc	=	0.073
M0d/Mc	=	0.095
N0DL/Mc	=	0.050
MrD/M1	=	0.181
NrDL/M1	=	0.097
M0D/M1	=	0.056
N0DL/M1	=	0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	101	102	101	102
Outplane Stress Summary	94	104	181	60

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	50	50	50	50
Inplane Stress Summary	39	58	270	-134

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	94	105	270	195

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	28	-28	28	-28
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50
Outplane Stress Summary	40	69	133	-7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-139	139	139	-139
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	101	102	101	102
Inplane Stress Summary	60	146	350	-120

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	60	146	350	121

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	44	45	44	45	44	45	44	45
Circ. Pl (SUS)	3	3	1	1	-2	-2	6	6



Circ. Q (SUS)	-22	22	46	-46	-15	15	49	-49
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-3	-3	7	7	4	4	0	0
Long. Q (SUS)	-22	22	56	-56	-16	16	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	44.0	45.0	44.0	45.0	44.0	45.0	44.0	45.0
Pm+Pl (SUS)	47.0	48.0	45.1	46.1	42.1	43.1	50.0	51.0
Pm+Pl+Q (Total)	29.7	70.8	92.2	27.7	27.5	58.5	99.7	22.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	138.00	Passed
Pm+Pl (SUS)	51.03	207.00	Passed
Pm+Pl+Q (TOTAL)	99.66	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	101	102	101	102	101	102	101	102
Circ. Pl (SUS)	-2	-2	18	18	2	2	12	12
Circ. Q (SUS)	-4	4	60	-60	-43	43	235	-235
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-2	-2	16	16	5	5	11	11
Long. Q (SUS)	-9	9	201	-201	-14	14	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	101.0	102.0	101.0	102.0	101.0	102.0	101.0	102.0
Pm+Pl (SUS)	99.0	100.0	119.0	120.0	103.1	104.1	113.1	114.1
Pm+Pl+Q (Total)	95.3	104.3	267.0	195.0	60.1	147.0	348.1	121.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	120.02	207.00	Passed
Pm+Pl+Q (TOTAL)	348.12	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
---------	------------------



Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	3	3	1	1	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-23	-23	27	27	-17	-17	21	21
Long. Q (SUS)	-73	73	163	-163	-53	53	143	-143
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	28.6	29.6	28.4	28.4	24.4	25.3	22.8	22.8
Pm+Pl+Q (Total)	104.8	55.1	191.3	140.9	77.3	38.6	166.4	128.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	29.61	177.00	Passed
Pm+Pl+Q (TOTAL)	191.34	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M1-24"
From: 30

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	1091.6666	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5450.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

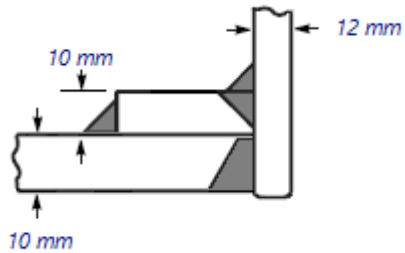
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	950.0001	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		170.2000	mm

Flange Class		150
Flange Grade		GR 1.4

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1-24''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9172 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 1.5576 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.9003 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1179.2000 mm
Parallel to Vessel Wall, opening length	d	589.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

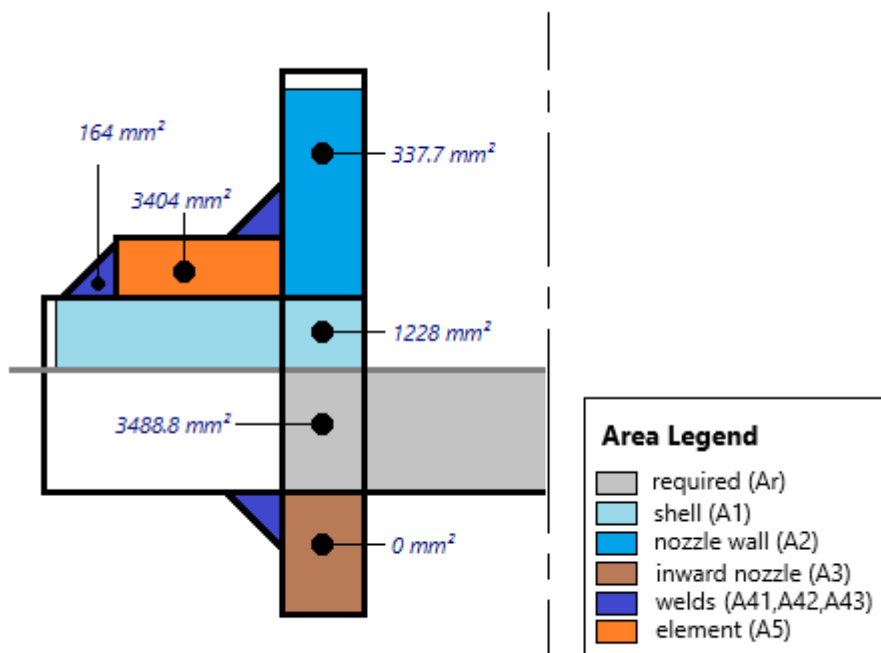


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3488.782	1581.789	NA
Area in Shell	A1	1228.018	1553.223	NA
Area in Nozzle Wall	A2	337.696	323.990	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	164.000	164.000	NA
Area in Element	A5	3404.000	3404.000	NA
TOTAL AREA AVAILABLE	Atot	5133.715	5445.213	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 785.5067 10.0000 mm
 Based on given Pad Diameter: 950.0001 5.1676 mm
 Based on Shell or Nozzle Thickness: 785.5067 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (589.6 \cdot 5.9172 \cdot 1.0 + 2 \cdot 10.0 \cdot 5.9172 \cdot 1.0 \cdot (1 - 1.0))$
 $= 3488.782 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 589.6(1.0 \cdot 8.0 - 1.0 \cdot 5.917) - 2 \cdot 10.0$
 $(1.0 \cdot 8.0 - 1.0 \cdot 5.9172) \cdot (1 - 1.0)$
 $= 1228.018 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 20.0) \cdot (10.0 - 1.56) \cdot 1.0$

= 337.696 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 8.0^2 \cdot 1.0$
 = 164.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(950.0 - 609.6) \cdot 10.0 \cdot 1.0$
 = 3404.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9003 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9172 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9172 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8351 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 10.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[10.331, max(7.9172, 3.5)]
 = 7.9172 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.9003, 7.9172)
 = 7.9172 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: D

Govrn. thk, tg = 12.0, tr = 1.558, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.156$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 10.0, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.74$, Temp. Reduction = 14 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
--	--------

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 10.0, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.74$, Temp. Reduction = 14 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
--	--------

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
--	--------

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 0.71/1.63 = 0.434

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3488.7817 - 1228.0182 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 8.0 - 5.9172)) \cdot 138)$
 $= 317706.88 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (337.6963 + 3404.0002 + 164.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 538940.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (337.6963 + 0.0 + 100.0 + (160.0)) \cdot 138$
 $= 82475.08 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (337.6963 + 0.0 + 164.0 + 3404.0002 + (160.0)) \cdot 138$
 $= 561018.44 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 950.0 \cdot 8.0 \cdot 0.49 \cdot 138$
 $= 807182. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 299.8) \cdot (12.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 909749. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 10.0 \cdot 0.74 \cdot 138$

= 977774. N

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 609.6 \cdot (10.0 - 2.0) \cdot 0.74 \cdot 138 \\ &= 782220. N \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (807183 + 909749) = 1716932 N$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (647445 + 977774 + 782220 + 0) = 2407440 N \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (807183 + 782220 + 0) = 1589402 N \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1716931 N	, must exceed W = 317706 N	or W1 = 538940 N
Path 2-2 = 2407439 N	, must exceed W = 317706 N	or W2 = 82475 N
Path 3-3 = 1589402 N	, must exceed W = 317706 N	or W3 = 561018 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 41.1280 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 641.1279 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %



Nozzle Layout or placement warning. Area for reinforcement may have been used that might not actually exist in this situation. Please check the nozzle geometry and placement carefully.

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Input, Nozzle Desc: N8-3''

From: 30

Pressure for Reinforcement Calculations	P	0.7067	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	2050.1003	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4550.00	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

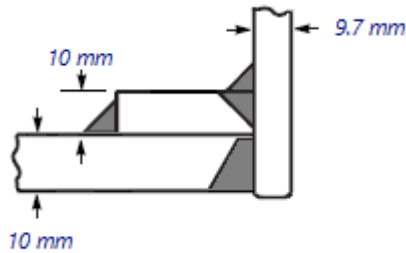
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	590.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	150.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		30.5500	mm

Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N8-3''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.71 \cdot 1152.0 / (138 \cdot 1.0 - 0.6 \cdot 0.71)$
 $= 5.9172 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 0.2656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6286 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.8618 mm
Parallel to Vessel Wall, opening length	d	73.4309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: N8-3''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.6286 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9172 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9172 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8351 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 6.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(7.9172, 3.5)]$
 $= 6.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.6286, 6.8)$
 $= 6.8000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 10.0, tr = 5.917, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
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Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 9.735, tr = 0.266, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $0.71/1.63 = 0.434$

Weld Size Calculations, Description: N8-3"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7345 mm
Intermediate Calc. for pad/shell Welds	TminPad	8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$4.0000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 600.8593 mm



Input, Nozzle Desc: M2-24"

From: 40

Pressure for Reinforcement Calculations	P	0.7123	MPa
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	2.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-40.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

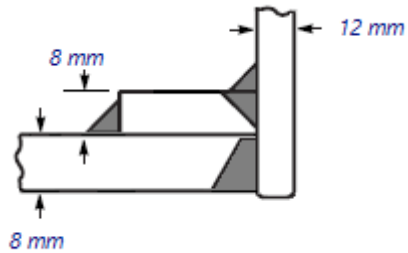
Corrosion Allowance	can	2.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	950.0001	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		170.2000	mm

Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M2-24''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.71 \cdot 2304.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.71)$
 $= 5.3457 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.71 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.71)$
 $= 1.5701 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4374 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1179.2000 mm
Parallel to Vessel Wall, opening length	d	589.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

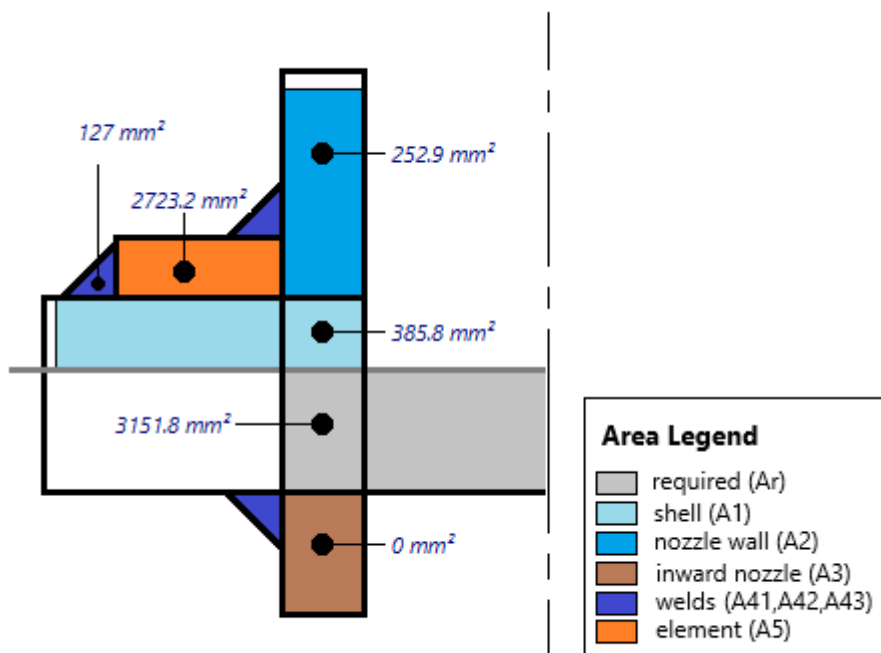


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3151.833	1738.271	NA
Area in Shell	A1	385.769	61.061	NA
Area in Nozzle Wall	A2	252.891	256.872	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	126.998	126.998	NA
Area in Element	A5	2723.200	2723.200	NA
TOTAL AREA AVAILABLE	Atot	3488.858	3168.131	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 907.8720 8.0000 mm
 Based on given Pad Diameter: 950.0001 7.0099 mm
 Based on Shell or Nozzle Thickness: 907.8720 8.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (589.6 \cdot 5.3457 \cdot 1.0 + 2 \cdot 10.0 \cdot 5.3457 \cdot 1.0 \cdot (1 - 1.))$
 $= 3151.833 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 589.6(1.0 \cdot 6.0 - 1.0 \cdot 5.346) - 2 \cdot 10.0$
 $(1.0 \cdot 6.0 - 1.0 \cdot 5.3457) \cdot (1 - 1.)$
 $= 385.769 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 15.0) \cdot (10.0 - 1.57) \cdot 1.$



= 252.891 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (91.0) • 1. + (0.0) • 1. + 6.0² • 1.0
 = 126.998 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (950.0 - 609.6) • 8.0 • 1.0
 = 2723.200 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5701 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9358 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9358 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.8329 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 10.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[10.331, max(7.9358, 3.5)]
 = 7.9358 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.5701, 7.9358)
 = 7.9358 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 8.0, c = 2.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.891, Temp. Reduction = 6 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 8.0, tr = 5.346, c = 2.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.891, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 8.0, tr = 5.346, c = 2.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.891, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -71 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 0.71/1.63 = 0.437

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	4.2420 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3151.8333 - 385.7688 + 2 \cdot 10.0 \cdot 1.0 \cdot 6.0 - 5.3457) \cdot 138)$
 $= 383490.12 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (252.8907 + 2723.2004 + 126.998 - 0.0 \cdot 1.0) \cdot 138$
 $= 428189.91 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (252.8907 + 0.0 + 99.9978 + (119.9974)) \cdot 138$
 $= 65252.71 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (252.8907 + 0.0 + 126.998 + 2723.2004 + (119.9974)) \cdot 138$
 $= 444748.16 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 647431. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 950.0 \cdot 6.0 \cdot 0.49 \cdot 138$
 $= 605387. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 299.8) \cdot (12.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 909730. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 8.0 \cdot 0.74 \cdot 138$
 $= 782220. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (8.0 - 2.0) \cdot 0.74 \cdot 138$
 $= 586665. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (605387 + 909730) = 1515116 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (647431 + 782220 + 586665 + 0) = 2016315 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (605387 + 586665 + 0) = 1192052 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1515116 N	, must exceed W = 383490 N	or W1 = 428189 N
Path 2-2 = 2016315 N	, must exceed W = 383490 N	or W2 = 65252 N
Path 3-3 = 1192051 N	, must exceed W = 383490 N	or W3 = 444748 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 22.4930 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.4931 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: